

PRACTICE PROBLEMS

1. Ratios are an input into which step in the financial statement analysis framework?
 - A. Process data
 - B. Collect input data
 - C. Analyze/interpret the processed data
2. Which phase in the financial statement analysis framework is *most likely* to involve producing updated reports and recommendations?
 - A. Follow-up
 - B. Analyze/interpret the processed data
 - C. Develop and communicate conclusions and recommendations
3. Which of the following *best* describes the role of financial statement analysis?
 - A. To provide information about a company's performance
 - B. To provide information about a company's changes in financial position
 - C. To form expectations about a company's future performance and financial position
4. The primary role of financial statement analysis is *best* described as:
 - A. providing information useful for making investment decisions.
 - B. evaluating a company for the purpose of making economic decisions.
 - C. using financial reports prepared by analysts to make economic decisions.
5. International Financial Reporting Standards are currently developed by which entity?
 - A. IFRS Foundation
 - B. International Accounting Standards Board
 - C. International Organization of Securities Commissions
6. US GAAP are currently developed by which entity?
 - A. Securities and Exchange Commission
 - B. Financial Accounting Standards Board
 - C. Public Company Accounting Oversight Board
7. A core objective of the International Organization of Securities Commissions is to:
 - A. eliminate systemic risk.
 - B. protect users of financial statements.

- C. ensure that markets are fair, efficient, and transparent.
8. Which of the following *best* describes why the notes that accompany the financial statements are required? The notes:
- A. permit flexibility in statement preparation.
 - B. standardize financial reporting across companies.
 - C. provide information necessary to understand the financial statements.
9. Accounting policies, methods, and estimates used in preparing financial statements are *most likely* to be found in the:
- A. auditor's report.
 - B. management commentary.
 - C. notes to the financial statements.
10. Information about management and director compensation is *most likely* to be found in the:
- A. auditor's report.
 - B. proxy statement.
 - C. earnings release.
11. Information about a company's objectives, strategies, and significant risks are *most likely* to be found in the:
- A. auditor's report.
 - B. management commentary.
 - C. notes to the financial statements.
12. What type of audit opinion is preferred when analyzing financial statements?
- A. Adverse
 - B. Qualified
 - C. Unqualified
13. An auditor determines that a company's financial statements are prepared in accordance with applicable accounting standards except with respect to inventory reporting. This exception is *most likely* to result in an audit opinion that is:
- A. adverse.
 - B. qualified.
 - C. unqualified.
14. An independent audit report is *most likely* to provide:
- A. absolute assurance about the accuracy of the financial statements.
 - B. reasonable assurance that the financial statements are fairly presented.

- C. a qualified opinion with respect to the transparency of the financial statements.
15. Interim financial reports released by a company are *most likely* to be:
- A. monthly.
 - B. unaudited.
 - C. unqualified.
16. Which of the following sources of information used by analysts is found outside a company's annual report?
- A. Auditor's report
 - B. Peer company analysis
 - C. Management discussion and analysis

SOLUTIONS

1. C is correct. Ratios are an output of the process information step but are an input into the analyze/interpret data step.
2. A is correct. The follow-up phase involves gathering information and repeating the analysis to determine whether it is necessary to update reports and recommendations.
3. C is correct. In general, analysts seek to examine the past and current performance and financial position of a company to form expectations about its future performance and financial position.
4. B is correct. The primary role of financial statement analysis is to use financial reports prepared by companies to evaluate their past, current, and potential performance and financial position for the purpose of making investment, credit, and other economic decisions.
5. B is correct. The International Accounting Standards Board (IASB) is currently charged with developing International Financial Reporting Standards.
6. B is correct. US Generally Accepted Accounting Principles are developed by the US Financial Accounting Standards Board (FASB).
7. C is correct. A core objective of IOSCO is to ensure that markets are fair, efficient, and transparent. The other core objectives are to reduce, not eliminate, systemic risk and to protect investors, not all users of financial statements.
8. C is correct. The notes provide information that is essential to understanding the information provided in the primary statements.
9. C is correct. The notes disclose choices in accounting policies, methods, and estimates.
10. B is correct. Disclosure of management compensation is typically included in the proxy statement. An earnings release is about corporate earnings, not what managers earn as compensation.
11. B is correct. These are components of management commentary.
12. C is correct. An unqualified opinion is a “clean” opinion and indicates that the financial statements present the company’s performance and financial position fairly in accordance with applicable accounting standards.
13. B is correct. A qualified audit opinion is one in which there is some scope limitation or exception to accounting standards. Exceptions are described in the audit report with additional explanatory paragraphs so that the analyst can determine the importance of the exception.
14. B is correct. The independent audit report provides reasonable assurance that the financial statements are fairly presented, meaning that there is a high probability that the audited financial statements are free from material error, fraud, or illegal acts that have a direct effect on the financial statements.
15. B is correct. Interim reports are typically provided semiannually or quarterly and require certain financial information, including unaudited financial statements

and an MD&A for the interim period covered by the report. Unqualified refers to a type of audit opinion.

16. B is correct. When performing financial statement analysis, analysts should review all company sources of information as well as information from external sources regarding the economy, the industry, the company, and peer (comparable) companies.

PRACTICE PROBLEMS

1. Under IFRS, income includes increases in economic benefits from:
 - A. increases in liabilities not related to owners' contributions.
 - B. enhancements of assets not related to owners' contributions.
 - C. increases in owners' equity related to owners' contributions.
2. Fairplay reported the information shown in Exhibit 1 related to the sale of its products during 2009, which was its first year of business:

Exhibit 1: Fairplay

Revenue	USD1,000,000
Returns of goods sold	USD100,000
Cash collected	USD800,000
Cost of goods sold	USD700,000

Under the accrual basis of accounting, how much net revenue would be reported on Fairplay's 2009 income statement?

- A. USD200,000
 - B. USD900,000
 - C. USD1,000,000
3. Apex Consignment sells items over the internet for individuals on a consignment basis. Apex receives the items from the owner, lists them for sale on the internet, and receives a 25 percent commission for any items sold. Apex collects the full amount from the buyer and pays the net amount after commission to the owner. Unsold items are returned to the owner after 90 days. During 2009, Apex had the following information:
 - Total sales price of items sold during 2009 on consignment was EUR2,000,000.
 - Total commissions retained by Apex during 2009 for these items was EUR500,000.

How much revenue should Apex report on its 2009 income statement?

 - A. EUR500,000
 - B. EUR2,000,000
 - C. EUR1,500,000
4. A company previously expensed the incremental costs of obtaining a contract. All else being equal, adopting the May 2014 IASB and FASB converged accounting standards on revenue recognition makes the company's profitability initially appear:
 - A. lower.

- B. unchanged.
 - C. higher.
5. Under IFRS, a loss from the destruction of property in a fire would most likely be classified as:
- A. continuing operations.
 - B. discontinued operations.
 - C. other comprehensive income.
6. A company chooses to change an accounting policy. This change requires that, if practical, the company restate its financial statements for:
- A. all prior periods.
 - B. current and future periods.
 - C. prior periods shown in a report.
7. For 2009, Flamingo Products had net income of USD1,000,000. At 1 January 2009, there were 1,000,000 shares outstanding. On 1 July 2009, the company issued 100,000 new shares for USD20 per share. The company paid USD200,000 in dividends to common shareholders. What is Flamingo's basic earnings per share for 2009?
- A. USD0.80
 - B. USD0.91
 - C. USD0.95
8. A company with no debt or convertible securities issued publicly traded common stock three times during the current fiscal year. Under both IFRS and US GAAP, the company's:
- A. basic EPS equals its diluted EPS.
 - B. capital structure is considered complex at year-end.
 - C. basic EPS is calculated by using a simple average number of shares outstanding.
9. For its fiscal year-end, Sublyme Corporation reported net income of USD200 million and a weighted average of 50,000,000 common shares outstanding. There are 2,000,000 convertible preferred shares outstanding that paid an annual dividend of USD5. Each preferred share is convertible into two shares of the common stock. The diluted EPS is *closest to*:
- A. USD3.52
 - B. USD3.65
 - C. USD3.70
10. For its fiscal year-end, Calvin Water Corporation (CWC) reported net income of USD12 million and a weighted average of 2,000,000 common shares outstanding. The company paid USD800,000 in preferred dividends and had 100,000 options

outstanding with an average exercise price of USD20. CWC's market price over the year averaged USD25 per share. CWC's diluted EPS is *closest* to:

- A. USD5.33
- B. USD5.54
- C. USD5.94

11. Laurelli Builders (LB) reported the financial data shown in Exhibit 1 for year-end 31 December:

Exhibit 1: Laurelli Builders

Common shares outstanding, 1 January	2,020,000
Common shares issued as stock dividend, 1 June	380,000
Warrants outstanding, 1 January	500,000
Net income	USD3,350,000
Preferred stock dividends paid	USD430,000
Common stock dividends paid	USD240,000

Which statement about the calculation of LB's EPS is *most* accurate?

- A. LB's basic EPS is USD1.12.
 - B. LB's diluted EPS is equal to or less than its basic EPS.
 - C. The weighted average number of shares outstanding is 2,210,000.
12. Cell Services Inc. (CSI) had 1,000,000 average shares outstanding during all of 2009. During 2009, CSI also had 10,000 options outstanding with exercise prices of USD10 each. The average stock price of CSI during 2009 was USD15. For purposes of computing diluted earnings per share, how many shares would be used in the denominator?
- A. 1,003,333
 - B. 1,006,667
 - C. 1,010,000
13. When calculating diluted EPS, which of the following securities in the capital structure increases the weighted average number of common shares outstanding without affecting net income available to common shareholders?
- A. Stock options
 - B. Convertible debt that is dilutive
 - C. Convertible preferred stock that is dilutive
14. Which statement is *most* accurate? A common size income statement:
- A. restates each line item of the income statement as a percentage of net income.

Practice Problems

- B.** allows an analyst to conduct cross-sectional analysis by removing the effect of company size.
- C.** standardizes each line item of the income statement but fails to help an analyst identify differences in companies' strategies.

SOLUTIONS

1. B is correct. Under IFRS, income includes increases in economic benefits from increases in assets, enhancement of assets, and decreases in liabilities.
2. B is correct. Net revenue is revenue for goods sold during the period less any returns and allowances, or USD1,000,000 minus USD100,000 = USD900,000.
3. A is correct. Apex is not the owner of the goods and should only report its net commission as revenue.
4. C is correct. Under the converged accounting standards, the incremental costs of obtaining a contract and certain costs incurred to fulfill a contract must be capitalized. If a company expensed these incremental costs in the years prior to adopting the converged standards, all else being equal, its profitability will appear higher under the converged standards.
5. A is correct. A fire may be infrequent, but it would still be part of continuing operations and reported in the profit and loss statement. Discontinued operations relate to a decision to dispose of an operating division.
6. C is correct. If a company changes an accounting policy, the financial statements for all fiscal years shown in a company's financial report are presented, if practical, as if the newly adopted accounting policy had been used throughout the entire period; this retrospective application of the change makes the financial results of any prior years included in the report comparable. Notes to the financial statements describe the change and explain the justification for the change.
7. C is correct. The weighted average number of shares outstanding for 2009 is 1,050,000. Basic earnings per share would be USD1,000,000 divided by 1,050,000, or USD0.95.
8. A is correct. Basic and diluted EPS are equal for a company with a simple capital structure. A company that issues only common stock, with no financial instruments that are potentially convertible into common stock has a simple capital structure. Basic EPS is calculated using the weighted average number of shares outstanding.

9. C is correct.

Diluted EPS

= (Net income)/(Weighted average number of shares outstanding + New common shares that would have been issued at conversion)

= USD200,000,000/[50,000,000 + (2,000,000 × 2)]

= USD3.70

The diluted EPS assumes that the preferred dividend is not paid and that the shares are converted at the beginning of the period.

10. B is correct. The formula to calculate diluted EPS is as follows:

Diluted EPS

$$= (\text{Net income} - \text{Preferred dividends}) / [\text{Weighted average number of shares outstanding} + (\text{New shares that would have been issued at option exercise} - \text{Shares that could have been purchased with cash received upon exercise}) \times (\text{Proportion of year during which the financial instruments were outstanding})]$$

The underlying assumption is that outstanding options are exercised, and then the proceeds from the issuance of new shares are used to repurchase shares already outstanding:

Proceeds from option exercise = $100,000 \times \text{USD}20 = \text{USD}2,000,000$

Shares repurchased = $\text{USD}2,000,000 / \text{USD}25 = 80,000$

The net increase in shares outstanding is thus $100,000 - 80,000 = 20,000$. Therefore, the diluted EPS for CWC = $(\text{USD}12,000,000 - \text{USD}800,000) / 2,020,000 = \text{USD}5.54$.

11. B is correct. LB has warrants in its capital structure; if the exercise price is less than the weighted average market price during the year, the effect of their conversion is to increase the weighted average number of common shares outstanding, causing diluted EPS to be lower than basic EPS. If the exercise price is equal to the weighted average market price, the number of shares issued equals the number of shares repurchased. Therefore, the weighted average number of common shares outstanding is not affected and diluted EPS equals basic EPS. If the exercise price is greater than the weighted average market price, the effect of their conversion is anti-dilutive. As such, they are not included in the calculation of basic EPS. LB's basic EPS is $\text{USD}1.22 [= (\text{USD}3,350,000 - \text{USD}430,000) / 2,400,000]$. Stock dividends are treated as having been issued retroactively to the beginning of the period.
12. A is correct. With stock options, the treasury stock method must be used. Under that method, the company would receive $\text{USD}100,000 (10,000 \times \text{USD}10)$ and would repurchase 6,667 shares ($\text{USD}100,000 / \text{USD}15$). The shares for the denominator would be:

Shares outstanding	1,000,000
Options exercises	10,000
Treasury shares purchased	(6,667)
Denominator	1,003,333

13. A is correct. When a company has stock options outstanding, diluted EPS is calculated as if the financial instruments had been exercised and the company had used the proceeds from the exercise to repurchase as many shares possible at the weighted average market price of common stock during the period. As a result, the conversion of stock options increases the number of common shares outstanding but has no effect on net income available to common shareholders. The conversion of convertible debt increases the net income available to common shareholders by the after-tax amount of interest expense saved. The conversion of convertible preferred shares increases the net income available to common shareholders by the amount of preferred dividends paid; the numerator becomes the net income.
14. B is correct. Common size income statements facilitate comparison across time periods (time-series analysis) and across companies (cross-sectional analysis) by stating each line item of the income statement as a percentage of revenue. The relative performance of different companies can be more easily assessed because

scaling the numbers removes the effect of size. A common size income statement states each line item on the income statement as a percentage of revenue. The standardization of each line item makes a common size income statement useful for identifying differences in companies' strategies.

PRACTICE PROBLEMS

1. All of the following are current assets *except*:
 - A. cash.
 - B. goodwill.
 - C. inventories.
2. The initial measurement of goodwill is *most likely* affected by:
 - A. an acquisition's purchase price.
 - B. the acquired company's book value.
 - C. the fair value of the acquirer's assets and liabilities.
3. For financial assets classified as trading securities, how are unrealized gains and losses reflected in shareholders' equity?
 - A. They are not recognized.
 - B. They flow through income into retained earnings.
 - C. They are a component of accumulated other comprehensive income.
4. For financial assets classified as available for sale, how are unrealized gains and losses reflected in shareholders' equity?
 - A. They are not recognized.
 - B. They flow through retained earnings.
 - C. They are a component of accumulated other comprehensive income.
5. For financial assets classified as held to maturity, how are unrealized gains and losses reflected in shareholders' equity?
 - A. They are not recognized.
 - B. They flow through retained earnings.
 - C. They are a component of accumulated other comprehensive income.
6. A company has total liabilities of GBP35 million and total stockholders' equity of GBP55 million. Total liabilities are represented on a vertical common-size balance sheet by a percentage *closest* to:
 - A. 35 percent.
 - B. 39 percent.
 - C. 64 percent.
7. Which of the following would an analyst *most likely* be able to determine from a common-size analysis of a company's balance sheet over several periods?
 - A. An increase or decrease in sales

- B. An increase or decrease in financial leverage
 - C. A more efficient or less efficient use of assets
8. Defining total asset turnover as revenue divided by average total assets, all else equal, impairment write-downs of long-lived assets owned by a company will *most likely* result in an increase for that company in:
- A. the debt-to-equity ratio but not the total asset turnover.
 - B. the total asset turnover but not the debt-to-equity ratio.
 - C. both the debt-to-equity ratio and the total asset turnover.
9. An investor concerned about a company's ability to meet its near-term obligations is *most likely* to calculate the:
- A. current ratio.
 - B. return on total capital.
 - C. financial leverage ratio.
10. The most stringent test of a company's liquidity is its:
- A. cash ratio.
 - B. quick ratio.
 - C. current ratio.
11. An investor worried about a company's long-term solvency would *most likely* examine its:
- A. current ratio.
 - B. return on equity.
 - C. debt-to-equity ratio.

The following information relates to questions 12-15

Consider the common-size balance sheets in Exhibit 1 for Company A, Company B, as well as the industry average.

Exhibit 1: Balance Sheet and Industry Average

	Company A	Company B	Industry Average
ASSETS			
Current assets			
Cash and cash equivalents	5	5	7

	Company A	Company B	Industry Average
Marketable securities	5	0	2
Accounts receivable, net	5	15	12
Inventories	15	20	16
Prepaid expenses	5	15	11
Total current assets	35	55	48
Property, plant, and equipment, net	40	35	37
Goodwill	25	0	8
Other assets	0	10	7
Total assets	100	100	100
LIABILITIES AND SHAREHOLDERS' EQUITY			
Current liabilities			
Accounts payable	10	10	10
Short-term debt	25	10	15
Accrued expenses	0	5	3
Total current liabilities	35	25	28
Long-term debt	45	20	28
Other non-current liabilities	0	10	7
Total liabilities	80	55	63
Total shareholders' equity	20	45	37
Total liabilities and shareholders' equity	100	100	100

12. Which statement is correct?

- A. Company A has below-average liquidity risk.
- B. Company B has above-average solvency risk.
- C. Company A has made one or more acquisitions.

13. The quick ratio for Company A is *closest* to:

- A. 0.43.
- B. 0.57.
- C. 1.00.

14. The financial leverage ratio for Company B is *closest* to:

- A. 0.55.
- B. 1.22.
- C. 2.22.

15. Which ratio indicates lower liquidity risk for Company A compared with Company B?

- A. Cash ratio

- B. Quick ratio
 - C. Current ratio
-

SOLUTIONS

1. B is correct. Goodwill is a long-term asset, and cash and inventories are current assets.
2. A is correct. Initially, goodwill is measured as the difference between the purchase price paid for an acquisition and the fair value of the acquired, not acquiring, company's net assets (identifiable assets less liabilities).
3. B is correct. For financial assets classified as trading securities, unrealized gains and losses are reported on the income statement and flow to shareholders' equity as part of retained earnings.
4. C is correct. For financial assets classified as available for sale, unrealized gains and losses are not recorded on the income statement and instead are part of *other* comprehensive income. Accumulated other comprehensive income is a component of shareholders' equity.
5. A is correct. Financial assets classified as held to maturity are measured at amortized cost. Gains and losses are recognized only when realized.
6. B is correct. Vertical common-size analysis involves stating each balance sheet item as a percentage of total assets. Total assets are the sum of total liabilities (GBP35 million) and total stockholders' equity (GBP55 million), or GBP90 million. Total liabilities are shown on a vertical common-size balance sheet as $(\text{GBP}35 \text{ million} \div \text{GBP}90 \text{ million}) \approx 39\%$.
7. B is correct. A common-size balance sheet analysis provides information about the composition of the balance sheet and it changes over time. As a result, it can provide information about an increase or decrease in a company's financial leverage.
8. C is correct. Impairment write-downs reduce equity in the denominator of the debt-to-equity ratio but do not affect debt, so the debt-to-equity ratio is expected to increase. Impairment write-downs reduce total assets but do not affect revenue. Thus, total asset turnover is expected to increase.
9. A is correct. The current ratio provides a comparison of assets that can be turned into cash relatively quickly and liabilities that must be paid within one year. The other ratios are more suited to evaluate longer-term concerns.
10. A is correct. The cash ratio determines how much of a company's near-term obligations can be settled with existing amounts of cash and marketable securities.
11. C is correct. The debt-to-equity ratio, a solvency ratio, is an indicator of financial risk.
12. C is correct. The presence of goodwill on Company A's balance sheet signifies that it has made one or more acquisitions in the past. The current, cash, and quick ratios are lower for Company A than for the sector average. These lower liquidity ratios imply above-average liquidity risk. The total debt, long-term debt-to-equity, debt-to-equity, and financial leverage ratios are lower for Company B than for the sector average. These lower solvency ratios imply below-average solvency risk.

Current ratio is $(35 \div 35) = 1.00$ for Company A, versus $(48 \div 28)$

= 1.71 for the sector average.

Cash ratio is $(5 + 5) \div 35 = 0.29$ for Company A, versus $(7 + 2) \div 28 = 0.32$ for the sector average.

Quick ratio is $(5 + 5 + 5) \div 35 = 0.43$ for Company A, versus $(7 + 2 + 12) \div 28 = 0.75$ for the sector average.

Total debt ratio is $(55 \div 100) = 0.55$ for Company B, versus $(63 \div 100) = 0.63$ for the sector average.

Long-term debt-to-equity ratio is $(20 \div 45) = 0.44$ for Company B, versus $(28 \div 37) = 0.76$ for the sector average.

Debt-to-equity ratio is $(55 \div 45) = 1.22$ for Company B, versus $(63 \div 37) = 1.70$ for the sector average.

Financial leverage ratio is $(100 \div 45) = 2.22$ for Company B, versus $(100 \div 37) = 2.70$ for the sector average.

13. A is correct. The quick ratio is defined as (Cash and cash equivalents + Marketable securities + receivables) $\div$ Current liabilities. For Company A, this calculation is $(5 + 5 + 5) \div 35 = 0.43$.
14. C is correct. The financial leverage ratio is defined as Total assets $\div$ Total equity. For Company B, total assets are 100 and total equity is 45; hence, the financial leverage ratio is $100 \div 45 = 2.22$.
15. A is correct. A higher cash ratio reflects lower liquidity risk. The cash ratio is defined as (Cash + Marketable securities) $\div$ Current liabilities. Company A's cash ratio, $(5 + 5) \div 35 = 0.29$, is higher than $(5 + 0) \div 25 = 0.20$ for Company B.

PRACTICE PROBLEMS

- Based on the information in Exhibit 1 for Pinkerly Inc., a fictitious company, what are the total adjustments that the company would make to net income in order to derive operating cash flow?

Exhibit 1: Pinkerly Inc.

Year Ended			
Income statement item	12/31/2018		
Net income	USD30 million		
Depreciation	USD7 million		
Balance sheet item	12/31/2017	12/31/2018	Change
Accounts receivable	USD15 million	USD30 million	USD15 million
Inventory	USD16 million	USD13 million	(USD3 million)
Accounts payable	USD10 million	USD20 million	USD10 million

- Add USD5 million
 - Add USD21 million
 - Subtract USD9 million
- When computing net cash flow from operating activities using the indirect method, an addition to net income is *most likely* to occur when there is a:
 - gain on the sale of an asset.
 - loss on the retirement of debt.
 - decrease in a deferred tax liability.
 - An analyst gathered the information in Exhibit 1 from a company's 2018 financial statements:

Exhibit 1: 2018 Financial Statement (US dollars, millions)

Balances as of Year Ended 31 December	2017	2018
Retained earnings	120	145
Accounts receivable	38	43
Inventory	45	48
Accounts payable	36	29

In 2018, the company declared and paid cash dividends of USD10 million and recorded depreciation expense in the amount of USD25 million. The company considers dividends paid a financing activity. The company's 2018 cash flow from operations (in USD millions) was *closest* to:

- A. 25.
 - B. 45.
 - C. 75.
4. Based on the information in Exhibit 1 for Star Inc., what are the total net adjustments that the company would make to net income to derive operating cash flow?

Exhibit 1: Star Inc.

Year Ended			
Income Statement			
Item	12/31/2018		
Net income	USD20 million		
Depreciation	USD2 million		
Balance Sheet Item	12/31/2017	12/31/2018	Change
Accounts receivable	USD25 million	USD22 million	(USD3 million)
Inventory	USD10 million	USD14 million	USD4 million
Accounts payable	USD8 million	USD13 million	USD5 million

- A. Add USD2 million
 - B. Add USD6 million
 - C. Subtract USD6 million.
5. In 2018, a company using US GAAP made cash payments of USD6 million for salaries, USD2 million for interest expense, and USD4 million for income taxes. Additional information for the company is provided in the Exhibit 1:

Exhibit 1: Cash Payments

(US dollars, millions)	2017	2018
Revenue	42	37
Cost of goods sold	18	16
Inventory	36	40
Accounts receivable	22	19
Accounts payable	14	12

Based only on the information in Exhibit 1, the company's operating cash flow for 2018 is *closest to*:

- A. USD6 million.
 - B. USD10 million.
 - C. USD14 million.
6. Green Glory Corp., a garden supply wholesaler, reported cost of goods sold for

the year of USD80 million. Total assets increased by USD55 million, including an increase of USD5 million in inventory. Total liabilities increased by USD45 million, including an increase of USD2 million in accounts payable. The cash paid by the company to its suppliers is most likely *closest* to:

- A. USD73 million.
 - B. USD77 million.
 - C. USD83 million.
7. Purple Fleur S.A., a retailer of floral products, reported cost of goods sold for the year of USD75 million. Total assets increased by USD55 million, but inventory declined by USD6 million. Total liabilities increased by USD45 million, and accounts payable increased by USD2 million. The cash paid by the company to its suppliers is most likely *closest* to:
- A. USD67 million.
 - B. USD79 million.
 - C. USD83 million.
8. White Flag, a women's clothing manufacturer, reported salaries expense of USD20 million. The beginning balance of salaries payable was USD3 million, and the ending balance of salaries payable was USD1 million. How much cash did the company pay in salaries?
- A. USD18 million
 - B. USD21 million
 - C. USD22 million
9. An analyst gathered the information in Exhibit 1 from a company's 2018 financial statements:

Exhibit 1: 2018 Financial Statements (US dollars, millions)

Year ended 31 December	2017	2018
Net sales	245.8	254.6
Cost of goods sold	168.3	175.9
Accounts receivable	73.2	68.3
Inventory	39.0	47.8
Accounts payable	20.3	22.9

Based only on the information in Exhibit 1, the company's 2018 statement of cash flows in the direct format would include amounts (in US dollars millions) for cash received from customers and cash paid to suppliers, respectively, that are *closest* to:

	Cash received from customers	Cash paid to suppliers
A.	249.7	169.7
B.	259.5	174.5
C.	259.5	182.1

10. Golden Cumulus Corp., a commodities trading company, reported interest expense of USD19 million and taxes of USD6 million. Interest payable increased by USD3 million, and taxes payable decreased by USD4 million over the period. How much cash did the company pay for interest and taxes?
- A. USD22 million for interest and USD10 million for taxes
- B. USD16 million for interest and USD2 million for taxes
- C. USD16 million for interest and USD10 million for taxes
11. The information in Exhibit 1 is extracted from Sweetfall Incorporated's financial statements.

Exhibit 1: Sweetfall Inc.

Income Statement		Balance Sheet Changes	
Revenue	USD56,800	Decrease in accounts receivable	USD1,324
Cost of goods sold	27,264	Decrease in inventory	501
Other operating expense	562	Increase in prepaid expense	6
Depreciation expense	2,500	Increase in accounts payable	1,063

The amount of cash Sweetfall Inc. paid to suppliers is:

- A. USD25,700.
- B. USD26,702.
- C. USD27,826.

SOLUTIONS

1. A is correct. To derive operating cash flow, the company would make the following adjustments to net income: add depreciation (a non-cash expense) of USD7 million; add the decrease in inventory of USD3 million; add the increase in accounts payable of USD10 million; and subtract the increase in accounts receivable of USD15 million. Total additions of USD20 million and total subtractions of USD15 million result in net total additions of USD5 million.
2. B is correct. An addition to net income is made when there is a loss on the retirement of debt, which is a non-operating loss. A gain on the sale of an asset and a decrease in deferred tax liability are both subtracted from net-income.
3. B is correct. All dollar amounts are in millions. Net income (NI) for 2018 is USD35. This amount is the increase in retained earnings, USD25, plus the dividends paid, USD10. Depreciation of USD25 is added back to net income, and the increases in accounts receivable, USD5, and in inventory, USD3, are subtracted from net income because they are uses of cash. The decrease in accounts payable is also a use of cash and, therefore, a subtraction from net income. Thus, cash flow from operations is $\text{USD}25 + \text{USD}10 + \text{USD}25 - \text{USD}5 - \text{USD}3 - \text{USD}7 = \text{USD}45$.
4. B is correct. To derive operating cash flow, the company would make the following adjustments to net income: Add depreciation (a non-cash expense) of USD2 million; add the decrease in accounts receivable of USD3 million; add the increase in accounts payable of USD5 million; and subtract the increase in inventory of USD4 million. Total additions would be USD10 million, and total subtractions would be USD4 million, which gives net additions of USD6 million.

5. A is correct.

Operating cash flows

= Cash received from customers – (Cash paid to suppliers + Cash paid to employees + Cash paid for other operating expenses + Cash paid for interest + Cash paid for income taxes)

Cash received from customers = Revenue + Decrease in accounts receivable

= $\text{USD}37 + \text{USD}3 = \text{USD}40$ million

Cash paid to suppliers

= Cost of goods sold + Increase in inventory + Decrease in accounts payable

= $\text{USD}16 + \text{USD}4 + \text{USD}2 = \text{USD}22$ million

Therefore, the company's operating cash flow = $\text{USD}40 - \text{USD}22 - \text{Cash paid for salaries} - \text{Cash paid for interest} - \text{Cash paid for taxes} = \text{USD}40 - \text{USD}22 - \text{USD}6 - \text{USD}2 - \text{USD}4 = \text{USD}6$ million.

6. C is correct. Cost of goods sold of USD80 million plus the increase in inventory of USD5 million equals purchases from suppliers of USD85 million. The increase in accounts payable of USD2 million means that the company paid USD83 million in cash (USD85 million minus USD2 million) to its suppliers.
7. A is correct. Cost of goods sold of USD75 million less the decrease in inventory of USD6 million equals purchases from suppliers of USD69 million. The increase

in accounts payable of USD2 million means that the company paid USD67 million in cash (USD69 million minus USD2 million).

8. C is correct. Beginning salaries payable of USD3 million plus salaries expense of USD20 million minus ending salaries payable of USD1 million equals USD22 million. Alternatively, the expense of USD20 million plus the USD2 million decrease in salaries payable equals USD22 million.
9. C is correct. Cash received from customers = Sales + Decrease in accounts receivable = $254.6 + 4.9 = 259.5$. Cash paid to suppliers = Cost of goods sold + Increase in inventory – Increase in accounts payable = $175.9 + 8.8 - 2.6 = 182.1$.
10. C is correct. Interest expense of USD19 million less the increase in interest payable of USD3 million equals interest paid of USD16 million. Tax expense of USD6 million plus the decrease in taxes payable of USD4 million equals taxes paid of USD10 million.
11. A is correct. The amount of cash paid to suppliers is calculated as follows:
= Cost of goods sold – Decrease in inventory – Increase in accounts payable
= $\text{USD}27,264 - \text{USD}501 - \text{USD}1,063$
= $\text{USD}25,700$.

PRACTICE PROBLEMS

1. One appropriate method of preparing a common-size cash flow statement is to show each line item:
 - A. of revenue and expense as a percentage of net revenue.
 - B. on the cash flow statement as a percentage of net revenue.
 - C. on the cash flow statement as a percentage of total cash outflows.
2. Which of the following is an appropriate method of computing free cash flow to the firm?
 - A. Add operating cash flows to capital expenditures and deduct after-tax interest payments.
 - B. Add operating cash flows to after-tax interest payments and deduct capital expenditures.
 - C. Deduct both after-tax interest payments and capital expenditures from operating cash flows.
3. The first step in cash flow statement analysis should be to:
 - A. evaluate consistency of cash flows.
 - B. determine operating cash flow drivers.
 - C. identify the major sources and uses of cash.
4. An analyst has calculated a ratio using as the numerator the sum of operating cash flow, interest, and taxes and as the denominator the amount of interest. What is this ratio, what does it measure, and what does it indicate?
 - A. This ratio is an interest coverage ratio, measuring a company's ability to meet its interest obligations and indicating a company's solvency.
 - B. This ratio is an effective tax ratio, measuring the amount of a company's operating cash flow used for taxes and indicating a company's efficiency in tax management.
 - C. This ratio is an operating profitability ratio, measuring the operating cash flow generated accounting for taxes and interest and indicating a company's liquidity.

SOLUTIONS

1. B is correct. An appropriate method to prepare a common-size cash flow statement is to show each line item on the cash flow statement as a percentage of net revenue. An alternative way to prepare a statement of cash flows is to show each item of cash inflow as a percentage of total inflows and each item of cash outflows as a percentage of total outflows.
2. B is correct. Free cash flow to the firm can be computed as operating cash flows plus after-tax interest expense less capital expenditures.
3. C is correct. An overall assessment of the major sources and uses of cash should be the first step in evaluating a cash flow statement.
4. A is correct. This ratio is an interest coverage ratio, measuring a company's ability to meet its interest obligations and indicating a company's solvency. This coverage ratio is based on cash flow information; another common formulation of the interest coverage ratio uses EBITDA based on the income statement as the numerator.

PRACTICE PROBLEMS

1. Carrying inventory at a value above its historical cost would *most likely* be permitted if:
 - A. the inventory was held by a producer of agricultural products.
 - B. financial statements were prepared using US GAAP.
 - C. the change resulted from a reversal of a previous write-down.
2. Eric's Used Book Store prepares its financial statements in accordance with IFRS. Inventory was purchased for GBP1 million and later marked down to GBP550,000. One of the books, however, was later discovered to be a rare collectible item, and the inventory is now worth an estimated GBP3 million. The inventory is *most likely* reported on the balance sheet at:
 - A. GBP550,000.
 - B. GBP1,000,000.
 - C. GBP3,000,000.
3. Fernando's Pasta purchased inventory and later wrote it down. The current net realizable value is higher than the value when written down. Fernando's inventory balance will *most likely* be:
 - A. higher if it complies with IFRS.
 - B. higher if it complies with US GAAP.
 - C. the same under US GAAP and IFRS.
4. A write-down of the value of inventory to its net realizable value will have a positive effect on the:
 - A. balance sheet.
 - B. income statement.
 - C. inventory turnover ratio.
5. Zimt AG uses the FIFO method, and Nutmeg Inc. uses the LIFO method. Compared to the cost of replacing the inventory, during periods of rising prices, the cost of sales reported by:
 - A. Zimt is too low.
 - B. Nutmeg is too low.
 - C. Nutmeg is too high.
6. Zimt AG uses the FIFO method, and Nutmeg Inc. uses the LIFO method. Compared to the cost of replacing the inventory, during periods of rising prices the ending inventory balance reported by:
 - A. Zimt is too high.
 - B. Nutmeg is too low.

- C. Nutmeg is too high.
7. Like many technology companies, TechnoTools operates in an environment of declining prices. Its reported profits will tend to be *highest* if it accounts for inventory using the:
- A. FIFO method.
 - B. LIFO method.
 - C. Weighted average cost method.
8. Compared to using the weighted average cost method to account for inventory, during a period in which prices are generally rising, the current ratio of a company using the FIFO method would *most likely* be:
- A. lower.
 - B. higher.
 - C. dependent upon the interaction with accounts payable.
9. Zimt AG wrote down the value of its inventory in 2017 and reversed the write-down in 2018. Compared to the ratios that would have been calculated if the write-down had never occurred, Zimt's reported that the 2017:
- A. current ratio was too high.
 - B. gross margin was too high.
 - C. inventory turnover was too high.
10. Zimt AG wrote down the value of its inventory in 2017 and reversed the write-down in 2018. Compared to the results the company would have reported if the write-down had never occurred, Zimt's reported that the 2018:
- A. profit was overstated.
 - B. cash flow from operations was overstated.
 - C. year-end inventory balance was overstated.
11. Compared to a company that uses the FIFO method, during periods of rising prices a company that uses the LIFO method will *most likely* appear more:
- A. liquid.
 - B. efficient.
 - C. profitable.
12. Nutmeg, Inc. uses the LIFO method to account for inventory. During years in which inventory unit costs are generally rising and in which the company purchases more inventory than it sells to customers, its reported gross profit margin will *most likely* be:
- A. lower than it would be if the company used the FIFO method.
 - B. higher than it would be if the company used the FIFO method.
 - C. about the same as it would be if the company used the FIFO method.

13. Compared to using the FIFO method to account for inventory, during periods of rising prices, a company using the LIFO method is *most likely* to report higher:
- A. net income.
 - B. cost of sales.
 - C. income taxes.
14. Carey Company reports under US GAAP, whereas Jonathan Company reports under IFRS. It is *least likely* that:
- A. Carey has reversed an inventory write-down.
 - B. Jonathan has reversed an inventory write-down.
 - C. Jonathan and Carey both use the FIFO inventory accounting method.

The following information relates to questions 15-21

Hans Annan, CFA, a food and beverage analyst, is reviewing Century Chocolate's inventory policies as part of his evaluation of the company. Century Chocolate, based in Switzerland, manufactures chocolate products and purchases and resells other confectionery products to complement its chocolate line. Annan visited Century Chocolate's manufacturing facility last year. He learned that cacao beans, imported from Brazil, represent the most significant raw material and that the work-in-progress inventory consists primarily of three items: roasted cacao beans, a thick paste produced from the beans (called chocolate liquor), and a sweetened mixture that needs to be "conched" to produce chocolate. On the tour, Annan learned that the conching process ranges from a few hours for lower-quality products to six days for the highest-quality chocolates. While there, Annan saw the facility's climate-controlled area where manufactured finished products (cocoa and chocolate) and purchased finished goods are stored prior to shipment to customers. After touring the facility, Annan had a discussion with Century Chocolate's CFO regarding the types of costs that were included in each inventory category.

Annan has asked his assistant, Joanna Kern, to gather some preliminary information regarding Century Chocolate's financial statements and inventories. He also asked Kern to calculate the inventory turnover ratios for Century Chocolate and another chocolate manufacturer for the most recent five years. Annan does not know Century Chocolate's most direct competitor, so he asks Kern to do some research and select the most appropriate company for the ratio comparison.

Kern reports back that Century Chocolate prepares its financial statements in accordance with IFRS. She tells Annan that the policy footnote states that raw materials and purchased finished goods are valued at purchase cost, whereas work in progress and manufactured finished goods are valued at production cost. Raw material inventories and purchased finished goods are accounted for using the FIFO method, and the weighted average cost method is used for other inventories. An allowance is established when the net realizable value of any inventory item is lower than the value calculated previously.

Kern provides Annan with the selected financial statements and inventory data for Century Chocolate. The ratio exhibit Kern prepared compares Century Chocolate's inventory turnover ratios to those of Gordon's Goodies, a US-based com-

pany. Annan returns the exhibit and tells Kern to select a different competitor that reports using IFRS rather than US GAAP. During this initial review, Annan asks Kern why she has not indicated whether Century Chocolate uses a perpetual or a periodic inventory system. Kern replies that she learned that Century Chocolate uses a perpetual system but did not include this information in her report because inventory values would be the same under either a perpetual or periodic inventory system. Annan tells Kern she is wrong and directs her to research the matter.

While Kern is revising her analysis, Annan reviews the most recent month's Cocoa Market Review from the International Cocoa Organization. He is drawn to the statement that "the ICCO daily price, averaging prices in both futures markets, reached a 29-year high in US dollar terms and a 23-year high in special drawing rights (SDRs) terms (the SDR unit comprises a basket of major currencies used in international trade: US dollar, euro, pound sterling, and yen)." Annan makes a note that he will need to factor the potential continuation of this trend into his analysis.

Exhibit 1: Century Chocolate Financial Statements

A. Century Chocolate Income Statements (millions of Swiss francs)		
For Years Ended 31 December	2018	2017
Sales	95,290	93,248
Cost of sales	−41,043	−39,047
Marketing, administration, and other expenses	−35,318	−42,481
Profit before taxes	18,929	11,720
Taxes	−3,283	−2,962
Profit for the period	15,646	8,758
B. Century Chocolate Balance Sheets (millions of Swiss francs)		
31 December	2018	
Cash, cash equivalents, and short-term investments	6,190	8,252
Trade receivables and related accounts, net	11,654	12,910
Inventories, net	8,100	7,039
Other current assets	2,709	2,812
Total current assets	28,653	31,013
Property, plant, and equipment, net	18,291	19,130
Other non-current assets	45,144	49,875
Total assets	92,088	100,018
Trade and other payables	10,931	12,299
Other current liabilities	17,873	25,265
Total current liabilities	28,804	37,564
Non-current liabilities	15,672	14,963
Total liabilities	44,476	52,527
Equity		
Share capital	332	341

B. Century Chocolate Balance Sheets (millions of Swiss francs)			
	31 December	2018	
Retained earnings and other reserves		47,280	47,150
Total equity		47,612	47,491
Total liabilities and shareholders' equity		92,088	100,018

C. Century Chocolate Supplementary Footnote Disclosures: Inventories (millions of Swiss francs)			
	31 December	2018	2017
Raw Materials		2,154	1,585
Work in Progress		1,061	1,027
Finished Goods		5,116	4,665
Total inventories before allowance		8,331	7,277
Allowance for write-downs to net realizable value		-231	-238
Total inventories net of allowance		8,100	7,039

D. Century Chocolate Inventory Record for Purchased Lemon Drops			
Date		Cartons	Per Unit Amount (Swiss francs)
	Beginning inventory	100	22
4 Feb 2018	Purchase	40	25
3 Apr 2018	Sale	50	32
23 Jul 2018	Purchase	70	30
16 Aug 2018	Sale	100	32
9 Sep 2018	Sale	35	32
15 Nov 2018	Purchase	100	28

E. Century Chocolate Net Realizable Value Information for Black Licorice Jelly Beans			
	2018	2017	
FIFO cost of inventory at 31 December (Swiss francs)	314,890	374,870	
Ending inventory at 31 December (kilograms)	77,750	92,560	
Cost per unit (Swiss francs)	4.05	4.05	
Net Realizable Value (Swiss francs per kilograms)	4.20	3.95	

15. The costs *least likely* to be included by the CFO as inventory are:
- A. storage costs for the chocolate liquor.
 - B. excise taxes paid to the government of Brazil for the cacao beans.
 - C. storage costs for chocolate and purchased finished goods awaiting shipment to customers.
16. What is the *most likely* justification for Century Chocolate's choice of inventory valuation method for its purchased finished goods?
- A. It is the preferred method under IFRS.
 - B. It allocates the same per unit cost to both cost of sales and inventory.

- C. Ending inventory reflects the cost of goods purchased most recently.
17. In Kern's comparative ratio analysis, the 2018 inventory turnover ratio for Century Chocolate is *closest* to:
- A. 5.07.
 - B. 5.42.
 - C. 5.55.
18. The *most accurate* statement regarding Annan's reasoning for requiring Kern to select a competitor that reports under IFRS for comparative purposes is that under US GAAP:
- A. fair values are used to value inventory.
 - B. the LIFO method is permitted to value inventory.
 - C. the specific identification method is permitted to value inventory.
19. Annan's statement regarding the perpetual and periodic inventory systems is most significant when which of the following costing systems is used?
- A. LIFO
 - B. FIFO
 - C. Specific identification
20. Ignoring any tax effect, the change in net realizable value of the black licorice jelly beans from 2017 to 2018 will *most likely* result in:
- A. an increase in gross profit of CHF7,775.
 - B. an increase in gross profit of CHF11,670.
 - C. no impact on cost of sales because under IFRS, write-downs cannot be reversed.
21. If the trend noted in the ICCO report continues and Century Chocolate plans to maintain constant or increasing inventory quantities, the *most likely* impact on Century Chocolate's financial statements related to its raw materials inventory will be:
- A. a cost of sales that more closely reflects current replacement values.
 - B. a higher allocation of the total cost of goods available for sale to cost of sales.
 - C. a higher allocation of the total cost of goods available for sale to ending inventory.
-

The following information relates to questions 22-27

Robert Groff, an equity analyst, is preparing a report on Crux Corp. As part of his report, Groff makes a comparative financial analysis between Crux and its two main competitors, Rolby Corp. and Mikko Inc. Crux and Mikko report under US GAAP and Rolby reports under IFRS.

Groff gathers information on Crux, Rolby, and Mikko. The relevant financial information on the three companies, and on the industry, is provided in Exhibit 1.

Exhibit 1: Selected Financial Information (millions of US dollars)

A. Balance Sheets and Income Statements			
	Crux	Rolby	Mikko
Inventory valuation method	LIFO	FIFO	LIFO
From the Balance Sheets			
As of 31 December 2018			
Inventory, gross	480	620	510
Valuation allowance	20	25	14
Inventory, net	460	595	496
Total debt	1,122	850	732
Total shareholders' equity	2,543	2,403	2,091
As of 31 December 2017			
Inventory, gross	465	602	401
Valuation allowance	23	15	12
Inventory, net	442	587	389
From the Income Statements			
Year Ended 31 December 2018			
Revenues	4,609	5,442	3,503
Cost of goods sold ^a	3,120	3,782	2,550
Net income	229	327	205
^a Charges included in cost of goods sold for inventory write-downs*	13	15	15
B. LIFO Reserve			
LIFO Reserve			
As of 31 December 2018	55	0	77
As of 31 December 2017	72	0	50
As of 31 December 2016	96	0	43
Tax Rate			
Effective tax rate	30%	30%	30%
C. Industry Information			
	2018	2017	2016
Raw materials price index	112	105	100
Finished goods price index	114	106	100

** This does not match the change in the inventory valuation allowance because the valuation allowance is reduced to reflect the valuation allowance attached to items sold and increased for additional necessary write-downs.*

To compare the financial performance of the three companies, Groff decides to convert LIFO figures into FIFO figures, and adjust figures to assume no valuation allowance is recognized by any company.

22. Crux's inventory turnover ratio computed as of 31 December 2018, after the adjustments suggested by Groff, is *closest* to:
- A. 5.67.
 - B. 5.83.
 - C. 6.13.
23. Rolby's net profit margin for the year ended 31 December 2018, after the adjustments suggested by Groff, is *closest* to:
- A. 6.01 percent.
 - B. 6.20 percent.
 - C. 6.28 percent.
24. Compared with its unadjusted debt-to-equity ratio, Mikko's debt-to-equity ratio as of 31 December 2018, after the adjustments suggested by Groff, is:
- A. lower.
 - B. higher.
 - C. the same.
25. Which company's gross profit margin would best reflect current costs of the industry?
- A. Crux.
 - B. Rolby.
 - C. Mikko.
26. Would Rolby's valuation method show a higher gross profit margin than Crux's under an inflationary, a deflationary, or a stable price scenario?
- A. Stable
 - B. Inflationary
 - C. Deflationary
27. Which group of ratios usually appears more favorable with an inventory write-down?
- A. Activity ratios
 - B. Solvency ratios
 - C. Profitability ratios
-

The following information relates to questions 28-37

ZP Corporation is a (hypothetical) multinational corporation headquartered in Japan that trades on numerous stock exchanges. ZP prepares its consolidated financial statements in accordance with US GAAP. Excerpts from ZP's 2018 annual report are presented below.

Exhibit 1: ZP Corporation Financial Statements

A. Consolidated Balance Sheets (millions in Japanese yen)				
	31 December	2017	2018	
Current Assets				
Cash and cash equivalents		JPY542,849	JPY814,760	
⋮		⋮	⋮	
Inventories		608,572	486,465	
⋮		⋮	⋮	
Total current assets		4,028,742	3,766,309	
⋮		⋮	⋮	
Total assets		JPY10,819,440	JPY9,687,346	
⋮		⋮	⋮	
Total current liabilities		JPY3,980,247	JPY3,529,765	
⋮		⋮	⋮	
Total long-term liabilities		2,663,795	2,624,002	
Minority interest in consolidated subsidiaries		218,889	179,843	
Total shareholders' equity		3,956,509	3,353,736	
Total liabilities and shareholders' equity		JPY10,819,440	JPY9,687,346	
B. Consolidated Statements of Income (millions in Japanese yen)				
	For the years ended 31 December	2016	2017	2018
Net revenues				
Sales of products		JPY7,556,699	JPY8,273,503	JPY6,391,240
Financing operations		425,998	489,577	451,950
		7,982,697	8,763,080	6,843,190
Cost and expenses				
Cost of products sold		6,118,742	6,817,446	5,822,805
Cost of financing operations		290,713	356,005	329,128
Selling, general and administrative		827,005	832,837	844,927
⋮		⋮	⋮	⋮
Operating income (loss)		746,237	756,792	−153,670
⋮		⋮	⋮	⋮
Net income		JPY548,011	JPY572,626	−JPY145,646

Exhibit 2: Excerpt from the 2018 Annual Report, Selected Disclosures**Management Discussion and Analysis of Financial Condition and Results of Operations**

Cost reduction efforts were offset by increased prices of raw materials, other production materials and parts. Inventories decreased during fiscal 2018 by JPY122.1 billion, or 20.1 percent, to JPY486.5 billion. This reflects the impacts of decreased sales volumes and fluctuations in foreign currency translation rates.

Management and Corporate Information

Risk	Industry	and	Business	Factors	Risks
The worldwide market for our products is highly competitive. ZP faces intense competition from other manufacturers in the respective markets in which it operates. Competition has intensified due to the worldwide deterioration in economic conditions. In addition, competition is likely to further intensify because of continuing globalization, possibly resulting in industry reorganization. Factors affecting competition include product quality and features, the amount of time required for innovation and development, pricing, reliability, safety, economy in use, customer service and financing terms. Increased competition may lead to lower unit sales and excess production capacity and excess inventory. This may result in a further downward price pressure.					
ZP's ability to adequately respond to the recent rapid changes in the industry and to maintain its competitiveness will be fundamental to its future success in maintaining and expanding its market share in existing and new markets.					

Notes to Consolidated Financial Statements

2.	Summary	of	significant	accounting	policies:
Inventories. Inventories are valued at cost, not in excess of market. Cost is determined on the “average-cost” basis, except for the cost of finished products carried by certain subsidiary companies, which is determined on a last-in, first-out (LIFO) basis. Inventories valued on the LIFO basis totaled JPY94,578 million and JPY50,037 million at 31 December 2017 and 2018, respectively. Had the FIFO basis been used for those companies using the LIFO basis, inventories would have been JPY10,120 million and JPY19,660 million higher than reported at 31 December 2017 and 2018, respectively.					
9.					Inventories:
Inventories consist of the following:					

31 December (millions in Japanese yen)	2017	2018
Finished goods	JPY 403,856	JPY 291,977
Raw materials	99,869	85,966
Work in process	79,979	83,890
Supplies and other	24,868	24,632
	JPY 608,572	JPY 486,465

28. The management discussion and analysis (MD&A) indicated that the prices of raw material, other production materials, and parts increased. Based on the inventory valuation methods described in Note 2, which inventory classification would *least accurately* reflect current prices?

- A. Raw materials
- B. Finished goods
- C. Work in process

29. According to Exhibit 2, the 2018 Annual Report, if the company had used the FIFO inventory valuation method instead of the LIFO inventory valuation method for a portion of its inventory, the 2017 inventory value would be *closest* to:
- A. JPY104,698 million.
 - B. JPY506,125 million.
 - C. JPY618,692 million.
30. If ZP had prepared its financial statement in accordance with IFRS, the inventory turnover ratio (using average inventory) for 2018 would be:
- A. lower.
 - B. higher.
 - C. the same.
31. Inventory levels decreased from 2017 to 2018 for all of the following reasons *except*:
- A. LIFO liquidation.
 - B. decreased sales volume.
 - C. fluctuations in foreign currency translation rates.
32. Which observation is *most likely* a result of looking only at the information reported in Exhibit 2, Note 9?
- A. Increased competition has led to lower unit sales.
 - B. There have been significant price increases in supplies.
 - C. Management expects a further downturn in sales during 2019.
33. Exhibit 2, Note 2, indicates that “inventories valued on the LIFO basis totaled JPY94,578 million and JPY50,037 million at 31 December 2017 and 2018, respectively.” Based on this, the LIFO reserve should *most likely*:
- A. increase.
 - B. decrease.
 - C. remain the same.
34. In Exhibit 2, the Industry and Business Risk excerpt states that, “Increased competition may lead to lower unit sales and excess production capacity and excess inventory. This may result in a further downward price pressure.” The downward price pressure could lead to inventory that is valued above current market prices or net realizable value. Any write-downs of inventory are *least likely* to have a significant effect on the inventory valued using:
- A. weighted average cost.
 - B. first-in, first-out (FIFO).
 - C. last-in, first-out (LIFO).

35. During periods of rising inventory unit costs, a company using the FIFO method

rather than the LIFO method will report a lower:

- A. current ratio.
 - B. inventory turnover.
 - C. gross profit margin.
36. Compared with a company that uses the FIFO method, during a period of rising unit inventory costs, a company using the LIFO method will *most likely* appear more:
- A. liquid.
 - B. efficient.
 - C. profitable.
37. In a period of declining inventory unit costs and constant or increasing inventory quantities, which inventory method is *most likely* to result in a higher debt-to-equity ratio?
- A. LIFO
 - B. FIFO
 - C. Weighted average cost
-

SOLUTIONS

1. A is correct. IFRS allow the inventories of producers and dealers of agricultural and forest products, agricultural produce after harvest, and minerals and mineral products to be carried at net realizable value even if above historical cost. (US GAAP treatment is similar.)
2. B is correct. Under IFRS, the reversal of write-downs is required if net realizable value increases. The inventory will be reported on the balance sheet at GBP1,000,000. The inventory is reported at the lower of cost or net realizable value.
3. A is correct. IFRS requires the reversal of inventory write-downs if net realizable values increase; US GAAP does not permit the reversal of write-downs. Therefore, Fernando's inventory balance would be higher under IFRS.
4. C is correct. Activity ratios (e.g., inventory turnover and total asset turnover) will be positively affected by a write-down to net realizable value because the asset base (denominator) is reduced. On the balance sheet, the inventory carrying amount is written down to its net realizable value and the loss in value (expense) is generally reflected on the income statement in the cost of goods sold, thus reducing gross profit, operating profit, and net income.
5. A is correct. Zimt uses the FIFO method, so its cost of sales represents units purchased at a (no longer available) lower price. Nutmeg uses the LIFO method, so its cost of sales is approximately equal to the current replacement cost of inventory.
6. B is correct. Nutmeg uses the LIFO method, and thus some of the inventory on the balance sheet was purchased at a (no longer available) lower price. Zimt uses the FIFO method, so the carrying value on the balance sheet represents the most recently purchased units and thus approximates the current replacement cost.
7. B is correct. In a declining price environment, the newest inventory is the lowest-cost inventory. In such circumstances, using the LIFO method (selling the newer, cheaper inventory first) will result in lower cost of sales and higher profit.
8. B is correct. In a rising price environment, inventory balances will be higher for the company using the FIFO method. Accounts payable are based on amounts due to suppliers, not the amounts accrued based on inventory accounting.
9. C is correct. The write-down reduced the value of inventory and increased cost of sales in 2017. The higher numerator and lower denominator mean that the inventory turnover ratio as reported was too high. Gross margin and the current ratio were both too low.
10. A is correct. The reversal of the write-down shifted the cost of sales from 2018 to 2017. The 2017 cost of sales was higher because of the write-down, and the 2018 cost of sales was lower because of the reversal of the write-down. As a result, the reported 2018 profits were overstated. Inventory balance in 2018 is the same because the write-down and reversal cancel each other out. Cash flow from operations is not affected by the non-cash write-down, but the higher profits in 2018 likely resulted in higher taxes and thus lower cash flow from operations.
11. B is correct. LIFO will result in lower inventory and higher cost of sales. Gross margin (a profitability ratio) will be lower, the current ratio (a liquidity ratio) will

be lower, and inventory turnover (an efficiency ratio) will be higher.

12. A is correct. LIFO will result in lower inventory and higher cost of sales in periods of rising costs compared to FIFO. Consequently, LIFO results in a lower gross profit margin than FIFO.
13. B is correct. During periods of rising prices, using the LIFO method increases cost of sales relative to the FIFO method, thus reducing profits and the taxes thereon.
14. A is correct. US GAAP does not permit inventory write-downs to be reversed.
15. C is correct. The storage costs for inventory awaiting shipment to customers are not costs of purchase, costs of conversion, or other costs incurred in bringing the inventories to their present location and condition and are not included in inventory. The storage costs for the chocolate liquor occur during the production process and are thus part of the conversion costs. Excise taxes are part of the purchase cost.
16. C is correct. The carrying amount of inventories under FIFO will more closely reflect current replacement values because inventories are assumed to consist of the most recently purchased items. FIFO is an acceptable, but not preferred, method under IFRS. Weighted average cost, not FIFO, is the cost formula that allocates the same per unit cost to both cost of sales and inventory.
17. B is correct. Inventory turnover = Cost of sales/Average inventory = $41,043/7,569.5 = 5.42$. Average inventory is $(8,100 + 7,039)/2 = 7,569.5$.
18. B is correct. For comparative purposes, the choice of a competitor that reports under IFRS is requested because LIFO is permitted under US GAAP.
19. A is correct. The carrying amount of the ending inventory may differ because the perpetual system will apply LIFO continuously throughout the year, liquidating layers as sales are made. Under the periodic system, the sales will start from the last layer in the year. Under FIFO, the sales will occur from the same layers regardless of whether a perpetual or periodic system is used. Specific identification identifies the actual products sold and remaining in inventory, and there will be no difference under a perpetual or periodic system.
20. A is correct. Gross profit will most likely increase by CHF7,775. The net realizable value has increased and now exceeds the cost. The write-down from 2017 can be reversed. The write-down in 2017 was 9,256 [$92,560 \times (4.05 - 3.95)$]. IFRS require the reversal of any write-downs for a subsequent increase in value of inventory previously written down. The reversal is limited to the lower of the subsequent increase or the original write-down. Only 77,750 kilograms remain in inventory; the reversal is $77,750 \times (4.05 - 3.95) = 7,775$. The amount of any reversal of a write-down is recognized as a reduction in cost of sales. This reduction results in an increase in gross profit.
21. C is correct. Using the FIFO method to value inventories when prices are rising will allocate more of the cost of goods available for sale to ending inventories (the most recent purchases, which are at higher costs, are assumed to remain in inventory) and less to cost of sales (the oldest purchases, which are at lower costs, are assumed to be sold first).
22. B is correct. Crux's adjusted inventory turnover ratio must be computed using cost of goods sold (COGS) under FIFO and excluding charges for increases in valuation allowances.

COGS (adjusted)

= COGS (LIFO method) – Charges included in cost of goods sold for inventory write-downs – Change in LIFO reserve

= USD3,120 million – 13 million – (55 million – 72 million)

= USD3,124 million

Note: Minus the change in LIFO reserve is equivalent to plus the decrease in LIFO reserve. The adjusted inventory turnover ratio is computed using average inventory under FIFO.

Ending inventory (FIFO) = Ending inventory (LIFO) + LIFO reserve

Ending inventory 2018 (FIFO) = USD480 + 55 = USD535

Ending inventory 2017 (FIFO) = USD465 + 72 = USD537

Average inventory = (USD535 + 537)/2 = USD536

Therefore, adjusted inventory turnover ratio equals:

Inventory turnover ratio = COGS/Average inventory = USD3,124/USD536 = 5.83

23. B is correct. Rolby's adjusted net profit margin must be computed using net income (NI) under FIFO and excluding charges for increases in valuation allowances.

NI (adjusted)

= NI (FIFO method) + Charges, included in cost of goods sold for inventory write-downs, after tax

= USD327 million + 15 million $\times$ (1 – 30%)

= USD337.5 million

Therefore, adjusted net profit margin equals:

Net profit margin = NI/Revenues = USD337.5/USD5,442 = 6.20%.

24. A is correct. Mikko's adjusted debt-to-equity ratio is lower because the debt (numerator) is unchanged and the adjusted shareholders' equity (denominator) is higher. The adjusted shareholders' equity corresponds to shareholders' equity under FIFO, excluding charges for increases in valuation allowances. Therefore, adjusted shareholders' equity is higher than reported (unadjusted) shareholders' equity.
25. C is correct. Mikko's and Crux's gross margin ratios would better reflect the current gross margin of the industry than Rolby because both use LIFO. LIFO recognizes as cost of goods sold the cost of the most recently purchased units; therefore, it better reflects replacement cost. However, Mikko's gross margin ratio best reflects the current gross margin of the industry because Crux's LIFO reserve is decreasing. This could reflect a LIFO liquidation by Crux which would distort gross profit margin.
26. B is correct. The FIFO method shows a higher gross profit margin than the LIFO method in an inflationary scenario, because FIFO allocates to cost of goods sold the cost of the oldest units available for sale. In an inflationary environment, these units are the ones with the lowest cost.

27. A is correct. An inventory write-down increases the cost of sales and reduces profit and reduces the carrying value of inventory and assets. This has a negative effect on profitability and solvency ratios. However, activity ratios appear positively affected by a write-down because the asset base, whether total assets or inventory (denominator), is reduced. The numerator, sales, in total asset turnover is unchanged, and the numerator, cost of sales, in inventory turnover is increased. Thus, turnover ratios are higher and appear more favorable as the result of the write-down.
28. B is correct. Finished goods least accurately reflect current prices because some of the finished goods are valued under the last-in, first-out (“LIFO”) basis. The costs of the newest units available for sale are allocated to cost of goods sold, leaving the oldest units (at lower costs) in inventory. ZP values raw materials and work in process using the weighted average cost method. While not fully reflecting current prices, some inflationary effect will be included in the inventory values.
29. C is correct. FIFO inventory = Reported inventory + LIFO reserve = JPY608,572 + 10,120 = JPY618,692. The LIFO reserve is disclosed in Note 2 of the notes to consolidated financial statements.
30. A is correct. The inventory turnover ratio would be lower. The average inventory would be higher under FIFO and cost of products sold would be lower by the increase in LIFO reserve. LIFO is not permitted under IFRS.
- Inventory turnover ratio = Cost of products sold ÷ Average inventory
- 2018 inventory turnover ratio as reported = 10.63
- = JPY5,822,805 / [(608,572 + 486,465) / 2].
- 2018 inventory turnover ratio adjusted to FIFO as necessary = 10.34
- = [JPY5,822,805 – (19,660 – 10,120)] / [(608,572 + 10,120 + 486,465 + 19,660) / 2].
31. A is correct. No LIFO liquidation occurred during 2018; the LIFO reserve increased from JPY10,120 million in 2017 to JPY19,660 million in 2018. Management stated in the MD&A that the decrease in inventories reflected the impact of decreased sales volumes and fluctuations in foreign currency translation rates.
32. C is correct. Finished goods and raw materials inventories are lower in 2018 when compared to 2017. Reduced levels of inventory typically indicate an anticipated business contraction.
33. B is correct. The decrease in LIFO inventory in 2018 would typically indicate that more inventory units were sold than produced or purchased. Accordingly, one would expect a liquidation of some of the older LIFO layers and the LIFO reserve to decrease. In actuality, the LIFO reserve *increased* from JPY10,120 million in 2017 to JPY19,660 million in 2018. This is not to be expected and is likely caused by the increase in prices of raw materials, other production materials, and parts of foreign currencies as noted in the MD&A. An analyst should seek to confirm this explanation.
34. C is correct. In a period of rising inventory costs, inventory valued using FIFO would have relatively higher values compared to inventory valued using LIFO. Thus, any mark downs of inventory values to NRV would have the least impact on inventories valued using the LIFO method as they are already conservatively valued.

35. B is correct. During a period of rising inventory costs, a company using the FIFO method will allocate a lower amount to cost of goods sold and a higher amount to ending inventory as compared with the LIFO method. The inventory turnover ratio is the ratio of cost of sales to ending inventory. A company using the FIFO method will produce a lower inventory turnover ratio as compared with the LIFO method. The current ratio (current assets/current liabilities) and the gross profit margin [gross profit/sales = (sales less cost of goods sold)/sales] will be higher under the FIFO method than under the LIFO method in periods of rising inventory unit costs.
36. B is correct. During a period of rising inventory prices, a company using the LIFO method will have higher cost of goods sold and a lower inventory compared with a company using the FIFO method. The inventory turnover ratio will be higher for the company using the LIFO method, thus making it appear more efficient. Current assets and gross profit margin will be lower for the company using the LIFO method, thus making it appear to be less liquid and less profitable.
37. B is correct. In an environment of declining inventory unit costs and constant or increasing inventory quantities, FIFO (in comparison with weighted average cost or LIFO) will have higher cost of goods sold and lower net income and inventory. Because both inventory and net income are lower, total equity is lower, resulting in a higher debt-to-equity ratio.

PRACTICE PROBLEMS

The following information relates to questions 1-3

An analyst is studying the impairment of the manufacturing equipment of WLP Corp., a UK-based corporation that reports under IFRS. He gathers the following information about the equipment:

Fair value	GBP16,800,000
Costs to sell	GBP800,000
Value in use	GBP14,500,000
Net carrying amount	GBP19,100,000

- Based on this information, the amount of impairment loss that WLP will need to report on its income statement related to the manufacturing equipment is *closest* to:
 - GBP2,300,000.
 - GBP3,100,000.
 - GBP4,600,000.
- Under IFRS, an impairment loss on a property, plant, and equipment asset is measured as the excess of the carrying amount over the asset's:
 - fair value.
 - recoverable amount.
 - undiscounted expected future cash flows.
- The impairment of intangible assets with finite lives affects:
 - only the balance sheet.
 - only the income statement.
 - both the balance sheet and the income statement.

The following information relates to questions 4-7

Melanie Hart, CFA, is a transportation analyst. Hart has been asked to write a research report on Altai Mountain Rail Company (AMRC). Like other companies in the railroad industry, AMRC's operations are capital intensive, with significant investments in long-lived tangible assets as property, plant, and equipment. In November of 2021, AMRC's board of directors hired a new team to manage the

company. In reviewing the company's 2022 annual report, Hart is concerned about some of the accounting choices that the new management has made. These choices differ from those of the previous management and from common industry practice. Hart has highlighted the following statements from the company's annual report:

Statement 1 "In 2022, AMRC spent significant amounts on track replacement and similar improvements. AMRC expensed rather than capitalized a significant proportion of these expenditures."

Statement 2 "AMRC uses the straight-line method of depreciation for both financial and tax reporting purposes to account for plant and equipment."

Statement 3 "In 2022, AMRC recognized an impairment loss of €50 million on a fleet of locomotives. The impairment loss was reported as 'other income' in the income statement and reduced the carrying amount of the assets on the balance sheet."

Exhibit 1 and Exhibit 2 contain AMRC's 2022 consolidated income statement and balance sheet. AMRC prepares its financial statements in accordance with International Financial Reporting Standards.

Exhibit 1: Consolidated Statement of Income

For the Years Ended 31 December	2022		2021	
	Euros millions	Revenues (%)	Euros millions	Revenues (%)
Operating revenues	2,600	100.0	2,300	100.0
Operating expenses				
Depreciation	(200)	(7.7)	(190)	(8.3)
Other operating expense	(1,590)	(61.1)	(1,515)	(65.9)
Total operating expenses	(1,790)	(68.8)	(1,705)	(74.2)
Operating income	810	31.2	595	25.8
Other income	(50)	(1.9)		0.0
Interest expense	(73)	(2.8)	(69)	(3.0)
Income before taxes	687	26.5	526	22.8
Income taxes	(272)	(10.5)	(198)	(8.6)
Net income	415	16	328	14.2

Exhibit 2: Consolidated Balance Sheet

As of 31 December	2022		2021	
	Euros millions	Assets (%)	Euros millions	Assets (%)
Assets				
Current assets	500	9.4	450	8.5
Property & equipment:				
Land	700	13.1	700	13.2
Plant & equipment	6,000	112.1	5,800	109.4

Total property & equipment	6,700	125.2	6,500	122.6
Accumulated depreciation	(1,850)	(34.6)	(1,650)	(31.1)
Net property & equipment	4,850	90.6	4,850	91.5
Total assets	5,350	100.0	5,300	100.0
Liabilities and Shareholders' Equity				
Current liabilities	480	9.0	430	8.1
Long-term debt	1,030	19.3	1,080	20.4
Other long-term provisions and liabilities	1,240	23.1	1,440	27.2
Total liabilities	2,750	51.4	2,950	55.7
Shareholders' equity				
Common stock and paid-in-surplus	760	14.2	760	14.3
Retained earnings	1,888	35.5	1,600	30.2
Other comprehensive losses	(48)	(0.9)	(10)	(0.2)
Total shareholders' equity	2,600	48.6	2,350	44.3
Total liabilities & shareholders' equity	5,350	100.0	5,300	100.0

4. With respect to Statement 1, which of the following is the *most likely* effect of management's decision to expense rather than capitalize these expenditures?
 - A. 2022 net profit margin is higher than if the expenditures had been capitalized.
 - B. 2022 total asset turnover is lower than if the expenditures had been capitalized.
 - C. Future profit growth will be higher than if the expenditures had been capitalized.
5. With respect to Statement 2, what would be the *most likely* effect in 2023 if AMRC were to switch to an accelerated depreciation method for both financial and tax reporting?
 - A. Net profit margin would increase.
 - B. Total asset turnover would decrease.
 - C. Cash flow from operating activities would increase.
6. With respect to Statement 3, what is the *most likely* effect of the impairment loss?
 - A. Net income in years prior to 2022 was likely understated.
 - B. Net profit margins in years after 2022 will likely exceed the 2022 net profit margin.
 - C. Cash flow from operating activities in 2022 was likely lower due to the impairment loss.
7. Based on Exhibit 1, the *best estimate* of the average remaining useful life of the

company's plant and equipment at the end of 2022 is:

- A. 20.75 years.
 - B. 24.25 years.
 - C. 30.00 years.
-

The following information relates to questions 8-13

Brian Jordan is interviewing for a junior equity analyst position at Orion Investment Advisors. As part of the interview process, Mary Benn, Orion's Director of Research, provides Jordan with information about two hypothetical companies, Alpha and Beta, and asks him to comment on the companies' financial statements and ratios. Both companies prepare their financial statements in accordance with International Financial Reporting Standards (IFRS) and are identical in all respects except for their accounting choices.

Jordan is told that, at the beginning of the current fiscal year, both companies purchased a major new computer system and began building new manufacturing plants for their own use. Alpha capitalized and Beta expensed the cost of the computer system; Alpha capitalized and Beta expensed the interest costs associated with the construction of the manufacturing plants.

Benn asks Jordan, "What was the impact of these decisions on each company's current fiscal year financial statements and ratios?"

Jordan responds, "Alpha's decision to capitalize the cost of its new computer system instead of expensing it results in lower net income, lower total assets, and higher cash flow from operating activities in the current fiscal year. Alpha's decision to capitalize its interest costs instead of expensing them results in a lower fixed asset turnover ratio and a higher interest coverage ratio."

Jordan is told that Alpha uses the straight-line depreciation method and Beta uses an accelerated depreciation method; both companies estimate the same useful lives for long-lived assets. Many companies in their industry use the units-of-production method.

Benn asks Jordan, "What are the financial statement implications of each depreciation method, and how do you determine a company's need to reinvest in its productive capacity?"

Jordan replies, "All other things being equal, the straight-line depreciation method results in the least variability of net profit margin over time, while an accelerated depreciation method results in a declining trend in net profit margin over time. The units-of-production can result in a net profit margin trend that is quite variable. I use a three-step approach to estimate a company's need to reinvest in its productive capacity. First, I estimate the average age of the assets by dividing net property, plant, and equipment by annual depreciation expense. Second, I estimate the average remaining useful life of the assets by dividing accumulated depreciation by depreciation expense. Third, I add the estimates of the average remaining useful life and the average age of the assets in order to determine the total useful life."

Jordan is told that at the end of the current fiscal year, Alpha revalued a manufacturing plant; this increased its reported carrying amount by 15 percent. There was no previous downward revaluation of the plant. Beta recorded an impairment loss on a manufacturing plant; this reduced its carrying value by 10 percent.

Benn asks Jordan “What was the impact of these decisions on each company’s current fiscal year financial ratios?”

Jordan responds, “Beta’s impairment loss increases its debt to total assets and fixed asset turnover ratios, and lowers its cash flow from operating activities. Alpha’s revaluation increases its debt to capital and return on assets ratios, and reduces its return on equity.”

8. Jordan’s response about the financial statement impact of Alpha’s decision to capitalize the cost of its new computer system is correct with respect to:
 - A. lower net income.
 - B. lower total assets.
 - C. higher cash flow from operating activities.
 9. Jordan’s response about the ratio impact of Alpha’s decision to capitalize interest costs is most likely *correct* with respect to the:
 - A. interest coverage ratio.
 - B. fixed asset turnover ratio.
 - C. interest coverage and fixed asset turnover ratios.
 10. Jordan’s response about the impact of the different depreciation methods on net profit margin is most likely *incorrect* with respect to:
 - A. accelerated depreciation.
 - B. straight-line depreciation.
 - C. units-of-production depreciation.
 11. Jordan’s response about his approach to estimating a company’s need to reinvest in its productive capacity is most likely *correct* regarding estimating the:
 - A. average age of the asset base.
 - B. total useful life of the asset base.
 - C. average remaining useful life of the asset base.
 12. Jordan’s response about the effect of Beta’s impairment loss is *incorrect* with respect to the impact on its:
 - A. debt to total assets.
 - B. fixed asset turnover.
 - C. cash flow from operating activities.
 13. Jordan’s response about the effect of Alpha’s revaluation is most likely *correct* with respect to the impact on its:
 - A. return on equity.
 - B. return on assets.
 - C. debt to capital ratio.
-

The following information relates to questions 14-19

A financial analyst at BETTO S.A. is analyzing the result of the sale of a vehicle for 85,000 Argentine pesos (ARP) on 31 December 2021. The analyst compiles the following information about the vehicle:

Acquisition cost of the vehicle	ARP100,000
Acquisition date	1 January 2019
Estimated residual value at acquisition date	ARP10,000
Expected useful life	9 years
Depreciation method	Straight-line

14. The result of the sale of the vehicle is *most likely*:
 - A. a loss of ARP 15,000.
 - B. a gain of ARP 15,000.
 - C. a gain of ARP 18,333.
15. CROCO S.p.A sells an intangible asset with a historical acquisition cost of EUR12 million and an accumulated amortization of EUR2 million and reports a loss on the sale of EUR3.2 million. Which of the following amounts is *most likely* the sale price of the asset?
 - A. EUR6.8 million
 - B. EUR8.8 million
 - C. EUR13.2 million
16. The gain or loss on a sale of a long-lived asset to which the revaluation model has been applied is *most likely* calculated using sales proceeds less:
 - A. carrying amount.
 - B. carrying amount adjusted for impairment.
 - C. historical cost net of accumulated depreciation.
17. According to IFRS, all of the following pieces of information about property, plant, and equipment must be disclosed in a company's financial statements and footnotes *except for*:
 - A. useful lives.
 - B. acquisition dates.
 - C. amount of disposals.
18. According to IFRS, all of the following pieces of information about intangible assets must be disclosed in a company's financial statements and footnotes *except for*:
 - A. fair value.
 - B. impairment loss.

- C. amortization rate.
19. Which of the following is a required financial statement disclosure for long-lived intangible assets under US GAAP?
- A. The useful lives of assets
 - B. The reversal of impairment losses
 - C. Estimated amortization expense for the next five fiscal years
-

The following information relates to questions 20-23

After reading the financial statements and footnotes of a company that reports under IFRS, an analyst identified the following three intangible assets:

- product patent expiring in 40 years;
 - copyright with no expiration date; and
 - goodwill acquired 2 years ago in a business combination.
20. Which of the three assets is an intangible asset with a finite useful life?
- A. Patent
 - B. Goodwill
 - C. Copyright
21. Intangible assets with finite useful lives *mostly* differ from intangible assets with infinite useful lives with respect to accounting treatment of:
- A. revaluation.
 - B. impairment.
 - C. amortization.
22. Costs incurred for intangible assets are generally expensed when they are:
- A. internally developed.
 - B. individually acquired.
 - C. acquired in a business combination.
23. Under US GAAP, when assets are acquired in a business combination, goodwill *most likely* arises from:
- A. contractual or legal rights.
 - B. assets that can be separated from the acquired company.
 - C. assets that are neither tangible nor identifiable intangible assets.
-

SOLUTIONS

1. B is correct. The impairment loss equals GBP3,100,000, calculated as:

$$\text{Impairment} = \max(\text{Fair value less costs to sell; Value in use}) - \text{Net carrying amount}$$

$$= \max(16,800,000 - 800,000; 14,500,000) - 19,100,000$$

$$= -3,100,000.$$
2. B is correct. Under IFRS, an impairment loss is measured as the excess of the carrying amount over the asset's recoverable amount. The recoverable amount is the higher of the asset's fair value less costs to sell and its value in use. Value in use is a discounted measure of expected future cash flows. Under US GAAP, assessing recoverability is separate from measuring the impairment loss. If the asset's carrying amount exceeds its undiscounted expected future cash flows, the asset's carrying amount is considered unrecoverable and the impairment loss is measured as the excess of the carrying amount over the asset's fair value.
3. C is correct. The carrying amount of the asset on the balance sheet is reduced by the amount of the impairment loss, and the impairment loss is reported on the income statement.
4. C is correct. Expensing, rather than capitalizing, an investment in long-term assets will result in higher expenses and lower net income and net profit margin in the current year. Future years' incomes will not include depreciation expense related to these expenditures. Consequently, year-to-year growth in profitability will be higher. If the expenses had been capitalized, the carrying amount of the assets would have been higher and the 2022 total asset turnover would have been lower.
5. C is correct. Switching to an accelerated depreciation method would increase depreciation expense and decrease income before taxes, taxes payable, and net income. Cash flow from operating activities would increase because of the resulting tax savings.
6. B is correct. 2022 net income and net profit margin are lower because of the impairment loss. Consequently, net profit margins in subsequent years are likely to be higher. An impairment loss suggests that insufficient depreciation expense was recognized in prior years, and net income was overstated in prior years. The impairment loss is a non-cash item and will not affect operating cash flows.
7. A is correct. The estimated average remaining useful life is 20.75 years, calculated as:

$$\text{Estimate of remaining useful life} = \text{Net plant and equipment} \div \text{Annual depreciation expense}$$

$$\text{Net plant and equipment} = \text{Plant \& equipment} - \text{Accumulated depreciation}$$

$$= \text{EUR}6,000 - \text{EUR}1,850 = \text{EUR}4,150$$

$$\text{Estimate of remaining useful life} = \text{Net P \& E} \div \text{Depreciation expense}$$

$$= \text{EUR}4,150 \div \text{EUR}200 = 20.75$$
8. C is correct. The decision to capitalize the costs of the new computer system results in higher cash flow from operating activities; the expenditure is reported as an outflow of investing activities. The company allocates the capitalized amount over the asset's useful life as depreciation or amortization expense rather than expensing it in the year of expenditure. Net income and total assets are higher in

the current fiscal year.

9. B is correct. Alpha's fixed asset turnover will be lower because the capitalized interest will appear on the balance sheet as part of the asset being constructed. Therefore, fixed assets will be higher and the fixed asset turnover ratio (total revenue/average net fixed assets) will be lower than if it had expensed these costs. Capitalized interest appears on the balance sheet as part of the asset being constructed instead of being reported as interest expense in the period incurred. However, the interest coverage ratio should be based on interest payments, not interest expense (earnings before interest and taxes/interest payments) and should be unchanged. To provide a true picture of a company's interest coverage, the entire amount of interest expenditure, both the capitalized portion and the expensed portion, should be used in calculating interest coverage ratios.
10. A is correct. Accelerated depreciation will result in an improving, not declining, net profit margin over time, because the amount of depreciation expense declines each year. Under straight-line depreciation, the amount of depreciation expense will remain the same each year. Under the units-of-production method, the amount of depreciation expense reported each year varies with the number of units produced.
11. B is correct. The estimated average total useful life of a company's assets is calculated by adding the estimates of the average remaining useful life and the average age of the assets. The average age of the assets is estimated by dividing accumulated depreciation by depreciation expense. The average remaining useful life of the asset base is estimated by dividing net property, plant, and equipment by annual depreciation expense.
12. C is correct. The impairment loss is a non-cash charge and will not affect cash flow from operating activities. The debt to total assets and fixed asset turnover ratios will increase, because the impairment loss will reduce the carrying amount of fixed assets and therefore total assets.
13. A is correct. In an asset revaluation, the carrying amount of the assets increases. The increase in the asset's carrying amount bypasses the income statement and is reported as other comprehensive income and appears in equity under the heading of revaluation surplus. Therefore, shareholders' equity will increase but net income will not be affected, so return on equity will decline. Return on assets and debt to capital ratios will also decrease.
14. B is correct. The result on the sale of the vehicle is a gain of 15,000, calculated as:
 Gain or loss on the sale = Sale proceeds – Carrying amount
 = Sale proceeds – (Acquisition cost – Accumulated depreciation)
 = 85,000 – {100,000 – [(100,000 – 10,000)/9 years) × 3 years]}
 = 15,000.
15. A is correct. Gain or loss on the sale = Sale proceeds – Carrying amount. Rearranging this equation, Sale proceeds = Carrying amount + Gain or loss on sale. Thus, Sale price = (12 million – 2 million) + (–3.2 million) = 6.8 million.
16. A is correct. The gain or loss on the sale of long-lived assets is computed as the sales proceeds minus the carrying amount of the asset at the time of sale. This is true under the cost and revaluation models of reporting long-lived assets. In the absence of impairment losses, under the cost model, the carrying amount will equal historical cost net of accumulated depreciation.
17. B is correct. IFRS do not require acquisition dates to be disclosed.

18. A is correct. IFRS do not require fair value of intangible assets to be disclosed.
19. C is correct. Under US GAAP, companies are required to disclose the estimated amortization expense for the next five fiscal years. Under US GAAP, there is no reversal of impairment losses. Disclosure of the useful lives—finite or indefinite and additional related details—is required under IFRS.
20. A is correct. A product patent with a defined expiration date is an intangible asset with a finite useful life. A copyright with no expiration date is an intangible asset with an indefinite useful life. Goodwill is no longer considered an intangible asset under IFRS and is considered to have an indefinite useful life.
21. C is correct. An intangible asset with a finite useful life is amortized, whereas an intangible asset with an indefinite useful life is not amortized. Rather, they are carried on the balance sheet at historical cost and are tested at least annually for impairment.
22. A is correct. The costs to internally develop intangible assets are generally expensed when incurred.
23. C is correct. Under both IFRS and US GAAP, if an item is acquired in a business combination and cannot be recognized as a tangible asset or identifiable intangible asset, it is recognized as goodwill. Under US GAAP, assets arising from contractual or legal rights and assets that can be separated from the acquired company are recognized separately from goodwill.

PRACTICE PROBLEMS

1. Which of the following is a potential drawback of compensating employees with stock options?
 - A. The grant may make employees adverse to risk.
 - B. The grant may make employees seek more risk.
 - C. Both of the above are potential drawbacks.:
2. Which of the following is typically an objective of a share-based compensation plan?
 - A. Attracting new employees
 - B. Maximizing executive compensation
 - C. Alignment of employees' interest with those of management
3. Which of the following statements is true?
 - A. Share-based compensation does not have to be treated as an expense, when no cash is exchanged.
 - B. Share-based compensation programs can take a variety of forms, including those that are equity-settled and those that are cash settled.
 - C. Employees will receive a benefit of the stock option as long as they work long enough for the option to vest.
4. Which of the following is a difference between a stock grant and a stock option grant?
 - A. Whereas the fair value of stock grants is usually based on the market value at the date of the grant, the fair value of option grants must be estimated.
 - B. Companies account for stock grants by allocating compensation expense over the employee service period, whereas compensation expense for stock options is expensed immediately.
 - C. Compensation expense is determined based on the market value of a share of stock on the grant date, whereas the measurement date for the value of an option is when the employee exercises the option.
5. Assume ABC Company, a fictional company provides the following disclosure about its stock compensation plans:

"The average fair value of shares granted was USD20.86, USD16.42, and USD17.80 in 2021, 2020, and 2019 respectively." If the company granted 18,000 shares, with a three-year vesting period in 2021, what is the annual compensation expense for the 2021 shares granted?

 - A. USD125,160
 - B. USD339,480
 - C. USD375,480

6. Assume XYZ Company discloses the following information in its Stock Compensation note: As of 31 December 2021, we had USD630 million of unrecognized compensation cost related to nonvested stock-based compensation awards granted under our plan. We expect to recognize this cost over a weighted average period of 3.2 years as stock-based compensation expense. What is the expected compensation expense in 2025?
- A. USD39 million
 - B. USD197 million
 - C. USD630 million
7. Beginning with fiscal year 2019, for leases with a term longer than one year, lessees report a right-to-use asset and a lease liability on the balance sheet:
- A. only for finance leases.
 - B. only for operating leases.
 - C. for both finance and operating leases.
8. For a lessor, the leased asset appears on the balance sheet and continues to be depreciated when the lease is classified as:
- A. a finance lease.
 - B. a sales-type lease.
 - C. an operating lease.
9. Under US GAAP, a lessor's reported revenues at lease inception will be *highest* if the lease is classified as:
- A. a sales-type lease.
 - B. an operating lease.
 - C. a direct financing lease.
10. Under both IFRS and US GAAP, a lessor in an operating lease recognizes:
- A. selling profit at lease inception.
 - B. a lease asset comprising the lease receivable and relevant residual value at lease inception.
 - C. lease receipts as income and related costs, including depreciation, as expenses over the lease term.
11. Compared with a finance lease, an operating lease:
- A. is similar to renting an asset.
 - B. is equivalent to the purchase of an asset.
 - C. has a term for the majority of the economic life of the leased asset.
12. Under US GAAP, a lessee's accounting for a long-term finance lease after incep-

tion will include:

- A. recognizing a single lease expense.
- B. recording depreciation expense on the right-of-use asset.
- C. increasing the balance of the lease liability by a portion of the lease payment.

13. A company enters into a finance lease agreement to acquire the use of an asset for three years with lease payments of EUR19,000,000 starting next year. The leased asset has a fair market value of EUR49,000,000 and the present value of the lease payments is EUR47,250,188. Based on this information, the value of the lease liability reported on the company's balance sheet at lease inception is *closest* to:

- A. EUR47,250,188.
- B. EUR49,000,000.
- C. EUR57,000,000.

14. Penben Corporation has a defined benefit pension plan. At 31 December, its pension obligation is EUR10 million and pension assets are EUR9 million. Under either IFRS or US GAAP, the reporting on the balance sheet would be *closest* to which of the following?

- A. EUR10 million is shown as a liability, and EUR9 million appears as an asset.
- B. EUR1 million is shown as a net pension obligation.
- C. Pension assets and obligations are not required to be shown on the balance sheet but only disclosed in footnotes.

15. The information below is associated with a company that offers its employees a defined benefit plan:

Fair value of fund's assets	USD1,500,000,000
Estimated pension obligations	USD2,600,000,000
Present value of estimated pension obligations	USD1,200,000,000

Based on this information, the company's balance sheet will present a net pension:

- A. asset of USD300,000,000.
- B. asset of USD1,400,000,000.
- C. liability of USD1,100,000,000.

SOLUTIONS

1. C is correct. Stock option grants may lead managers to be either risk adverse or have the opposite effect (i.e., encourage excessive risk taking). Therefore, B and C are both potential drawbacks.
2. The correct answer is A. The objectives of employee compensation plans include attracting new employees, retaining and motivating existing employees and aligning employee interests with those of shareholders. Answer B, maximizing executive compensation is not typically an objective of a share-based compensation programs. Answer C is not correct because an objective is to align employee interests with shareholders, not necessarily management.
3. The correct answer is B. There are numerous types of share-based compensation programs; some result in the issuance of shares to employees, and others result in a cash payment based on the value of company shares. Answer A is not correct because even if no cash is exchanged at the time of grant, the compensation expense is recognized based on the fair value of the grant. Answer C is not correct because, in some cases, employees may not receive the benefit of a stock option. For example, if the stock option expires when the exercise price exceeds the market value of the company's stock, the recipient may not benefit from the grant.
4. A is correct. The compensation for a stock grant is based on the market value at the date of the stock grant. For a stock option, the value is not definitively known and must be estimated. Answer B is not correct because companies account for both stock grants and option grants by allocating the value of the grant over the service period (often the vesting period). Answer C is not correct because for both a share grant and an option grant, the value of the grant is determined based on the date of the grant.
5. The correct answer is A, calculated as follows: 18,000 shares $\times$ by average fair value at grant date of USD20.86 = USD375,480 total compensation. Divide this amount by the three-year vesting period, and the result is USD125,160 annual compensation expense.
6. The correct answer is A. USD39 million calculated as follows: USD630 million/3.2 years = USD197 million per year for years 2022, 2023, and 2024. The amount remaining in 2025 would be USD39 million: USD630 – USD197 – USD197 – USD197 = USD39 million.
7. C is correct. Beginning with fiscal year 2019, lessees report a right-of-use asset and a lease liability for all leases longer than one year. An exception under IFRS exists for leases when the underlying asset is of low value.
8. C is correct. When a lease is classified as an operating lease, the underlying asset remains on the lessor's balance sheet. The lessor will record a depreciation expense that reduces the asset's value over time.
9. A is correct. A sales-type lease treats the lease as a sale of the asset, and revenue is recorded at the time of sale equal to the value of the leased asset. Under a direct financing lease, only interest income is reported as earned. Under an operating lease, revenue from lease receipts is reported when collected.
10. C is correct. Lessor accounting for an operating lease under US GAAP is similar to that under IFRS: Over the lease term, the lessor recognizes lease receipts

as income and recognizes related costs, including depreciation of the leased asset, as expenses. Under IFRS, at inception of a finance lease—not an operating lease—the lessor derecognizes the underlying leased asset and recognizes a lease asset comprising the lease receivable and relevant residual value. Further, an IFRS-reporting lessor will recognize selling profit at the beginning of all leases that are not classified as operating leases. In contrast, a US GAAP-reporting lessor will recognize selling profit only on sales-type leases at the beginning of the lease term.

11. A is correct. An operating lease is an agreement that allows the lessee to use an asset for a period of time. Thus, an operating lease is similar to renting an asset, whereas a finance lease is equivalent to the purchase of an asset by the lessee that is directly financed by the lessor.
12. B is correct. A lessee's accounting for a long-term finance lease under US GAAP and after lease inception includes recording depreciation expense on the right-of-use asset, recognizing interest expense on the lease liability, and reducing the balance of the lease liability for the portion of the lease payments that represents repayment of the lease liability. A lessee's accounting for an operating lease under US GAAP and after lease inception will recognize a single lease expense, which is a straight-line allocation of the cost of the lease over its term.
13. A is correct. Under the revised reporting standards under IFRS and US GAAP, a lessee must recognize an asset and a lease liability at inception of each of its leases (with an exception for short-term leases). The lessee reports a right-of-use (ROU) asset and a lease liability, calculated essentially as the present value of fixed lease payments, on its balance sheet. Thus, at lease inception, the company will record a lease liability on the balance sheet of EUR47,250,188.
14. B is correct. The company will report a net pension obligation of EUR1 million equal to the pension obligation (EUR10 million) less the plan assets (EUR9 million).
15. A is correct. A company that offers a defined benefit plan makes payments into a pension fund and the retirees are paid from the fund. The payments that a company makes into the fund are invested until they are needed to pay retirees. If the fair value of the fund's assets is higher than the present value of the estimated pension obligation, the plan has a surplus and the company's balance sheet will reflect a net pension asset. Because the fair value of the fund's assets is USD1,500,000,000 and the present value of estimated pension obligations is USD1,200,000,000, the company will present a net pension asset of USD300,000,000 on its balance sheet.

PRACTICE PROBLEMS

- In the current year, Michaels Company has a carrying amount of USD3,500,000 and tax base of USD5,000,000 for accounts receivable. Michaels will most likely recognize:
 - a deferred tax asset.
 - a deferred tax liability.
 - no deferred tax asset or liability.
- James Company has received USD500,000 of tax credits from the recent installation of solar panels that will directly reduce their taxes. Which of the following *best* describes these tax credits?
 - Permanent difference
 - Taxable temporary difference
 - Deductible temporary difference
- Please use the selected data in Exhibit 1 for the Samuels Corporation.

Exhibit 1: Selected Data for Samuels Company (US dollar millions)

	Year 3	Year 2	Year 1
Equipment value for accounting purposes (<i>carrying amount</i>) (depreciation of USD1,000/year)	USD7,000	USD8,000	USD9,000
Equipment value for tax purposes (<i>tax base</i>) (depreciation of USD1,429/year)	USD5,714	USD7,143	USD8,571

Assuming a 35 percent tax rate and the selected data below for the Samuels Company, the company's deferred tax liability in Year 3 is *closest* to:

- USD450.
 - USD750.
 - USD900.
- Which of the following is added to income tax payable to determine the company's income tax expense as reported on the income statement?
 - Deferred tax assets
 - Deferred tax liabilities
 - Changes in deferred tax assets and liabilities
 - Jamison Corp. is domiciled in the United States and has significant operations in the United Kingdom and Australia. The statutory tax rates are 21 percent in the United States, 19 percent in the United Kingdom, and 30 percent in Australia. The company generates Profit before tax of USD2,000,000 in the United States,

USD500,000 in the United Kingdom, and USD750,000 in Australia. There are no other differences between Jamison's effective and statutory tax rates. Jamison's combined effective tax rate is closest to:

- A. 21.0 percent.
 - B. 22.8 percent.
 - C. 23.3 percent.
6. Which of the following statements about tax rates is correct?
- A. The effective tax rate is typically used for forecasting cash flows.
 - B. The cash tax rate is relevant for projecting earnings on the income statement.
 - C. A company's income tax expense equals the sum of current taxes plus the change in deferred tax assets and liabilities.

The following information relates to questions 7-8

Please use the selected disclosure data in Exhibit 1 and Exhibit 2 for the Marcy Corporation.

Note I: Income Taxes

The components of earnings before income taxes are as shown in Exhibit 1:

Exhibit 1: Earnings before Income Taxes (US dollars thousands)

	Year 3	Year 2	Year 1
Earnings before income taxes:			
United States	USD117,758	USD107,053	USD97,321
Foreign	57,526	52,296	47,542
Total	USD175,284	USD159,349	USD144,863

The components of the provision for income taxes are as shown in Exhibit 2:

Exhibit 2: Provision for Income Taxes (US dollars thousands)

	Year 3	Year 2	Year 1
Income taxes			
Current:			
Federal	USD34,739	USD34,257	USD31,143
Foreign	14,382	13,074	17,591
	USD49,121	USD47,331	USD48,734
Deferred:			
Federal	(USD6,524)	(USD6,002)	(USD5,325)

	Year 3	Year 2	Year 1
Foreign	389	305	262
	(6,135)	(5,697)	(5,063)
Total	USD42,986	USD41,634	USD43,671

7. Marcy's effective tax rate was lowest in:
- A. Year 1.
 - B. Year 2.
 - C. Year 3.
8. Relative to Marcy's effective tax rate on foreign income, the company's effective tax rate on US income was:
- A. lower in each year presented.
 - B. higher in each year presented.
 - C. higher in some periods and lower in others.

SOLUTIONS

1. A is correct. Because the carrying amount is less than the tax base for this asset, this difference is a temporary difference that will result in a deferred tax asset. B is incorrect because a deferred tax liability would apply if the carrying amount was greater than the asset base. C is incorrect because this is not a permanent difference thus there will be either a deferred tax asset or deferred tax liability.
2. A is correct. Permanent differences are differences between tax laws and accounting standards that will not be reversed at some future date. Because they will not be reversed at a future date, these differences do not give rise to deferred tax. These items include tax credits for expenditures that directly reduce taxes, such as tax credits related to the purchase of solar power. B is incorrect because taxable temporary differences result in the recognition of deferred tax liabilities. C is incorrect because deductible temporary differences result in a deferred tax asset.
3. A is correct. USD450 is calculated as: $(\text{USD}7,000 - \text{USD}5,714) \times 0.35 = \text{USD}450$. B is incorrect because it incorrectly sums the deferred tax liabilities from Years 2 and 3: $(\text{USD}7,000 - \text{USD}5,714) \times 0.35 + (\text{USD}8,000 - \text{USD}7,143) \times 0.35 = \text{USD}750$. C is incorrect because it incorrectly sums the deferred tax liabilities from Years 1, 2 and 3: $(\text{USD}7,000 - \text{USD}5,714) \times 0.35 + (\text{USD}8,000 - \text{USD}7,143) \times 0.35 + (\text{USD}9,000 - \text{USD}8,571) \times 0.35 = \text{USD}900$.
4. C is correct. The changes in deferred tax assets and liabilities are added to income tax payable to determine the company's income tax expense (or credit) as it is reported on the income statement. A and B are incorrect because it is the changes in deferred tax assets and liabilities that are added to income tax payable.
5. B is correct. The combined effective tax rate is calculated as:

Country	Taxable Income	Statutory Rate	Taxes
U.S.	USD2,000,000	21%	USD420,000
U.K.	500,000	19	95,000
Australia	750,000	30	225,000
Total	USD3,250,000		740,000
Effective tax rate:		740,000/3,250,000	22.8%

The effective tax rate is a blend of the different tax rates of the countries in which the activities take place in relation to the profit generated in each country.

A is incorrect because 21.0 percent is the statutory tax rate in the US and does not incorporate statutory tax rates in the United Kingdom and Australia. C is incorrect because 23.3 percent is the simple average of all three statutory tax rates.

6. C is correct. A company's income tax expense equals the sum of current taxes (i.e., the amount currently payable) plus the change in deferred tax assets and liabilities. A is incorrect because the cash tax rate is typically used for forecasting cash flows. B is incorrect because the effective tax rate is relevant for projecting earnings on the income statement.
7. C is correct. The effective tax rate of 24.5 percent ($\text{USD}42,986/\text{USD}175,284$) in Year 3 was lower than the effective tax rates in Year 1 and Year 2. A is incorrect

because its effective tax rate of 30.1 percent is higher than that of Year 3. B is incorrect because its effective tax rate of 26.1 percent is higher than that of Year 3.

8. C is correct. In Year 1, the effective tax rate on foreign operations was 37.6 percent $[(USD17,591 + USD262)/USD47,542]$, and the effective US tax rate was $[(USD31,143 - USD5,325)/USD97,321] = 26.5$ percent. In Year 2, the effective tax rate on foreign operations was 25.6 percent, and the US rate was 26.4 percent. In Year 3, the foreign rate was 25.7 percent, and the US rate was 24.0 percent.

PRACTICE PROBLEMS

1. In contrast to earnings quality, financial reporting quality *most likely* pertains to:
 - A. sustainable earnings.
 - B. relevant information.
 - C. adequate return on investment.
2. The information provided by a low-quality financial report will *most likely*:
 - A. decrease company value.
 - B. indicate earnings are not sustainable.
 - C. impede the assessment of earnings quality.
3. To properly assess a company's past performance, an analyst requires:
 - A. high earnings quality.
 - B. high financial reporting quality.
 - C. both high earnings quality and high financial reporting quality.
4. Low quality earnings *most likely* reflect:
 - A. low-quality financial reporting.
 - B. company activities which are unsustainable.
 - C. information that does not faithfully represent company activities.
5. Earnings that result from non-recurring activities *most likely* indicate:
 - A. lower-quality earnings.
 - B. biased accounting choices.
 - C. lower-quality financial reporting.
6. Which attribute of financial reports would *most likely* be evaluated as optimal in the financial reporting spectrum?
 - A. Conservative accounting choices
 - B. Sustainable and adequate returns
 - C. Emphasized pro forma earnings measures
7. Financial reports of the lowest level of quality reflect:
 - A. fictitious events.
 - B. biased accounting choices.
 - C. accounting that is non-compliant with GAAP.

8. When earnings are increased by deferring research and development (R&D) investments until the next reporting period, this choice is considered:
 - A. non-compliant accounting.
 - B. earnings management as a result of a real action.
 - C. earnings management as a result of an accounting choice.
9. A high-quality financial report may reflect:
 - A. earnings smoothing.
 - B. low earnings quality.
 - C. understatement of asset impairment.
10. If a particular accounting choice is considered aggressive in nature, then the financial performance for the reporting period would *most likely*:
 - A. be neutral.
 - B. exhibit an upward bias.
 - C. exhibit a downward bias.
11. Conservative accounting choices will most likely lead to:
 - A. decreased reported earnings in later periods.
 - B. increased reported earnings in the period under review.
 - C. increased debt reported on the balance sheet at the end of the current period.
12. Which of the following is *most likely* to be considered a potential benefit of accounting conservatism?
 - A. A reduction in litigation costs
 - B. Less biased financial reporting
 - C. An increase in current period reported performance
13. Which of the following statements *most likely* describes a situation that would motivate a manager to issue low-quality financial reports? The manager has:
 - A. increased the market share of products significantly.
 - B. earned compensation that is linked to stock price performance.
 - C. brought the company's profitability to a level higher than competitors.
14. Which of the following concerns would *most likely* motivate a manager to make conservative accounting choices?
 - A. Attention to future career opportunities
 - B. Debt covenant violation risk in the current period
 - C. Unexpected strength in the business environment

15. Which of the following conditions *best* explains why a company's manager would obtain legal, accounting, and board level approval prior to issuing low-quality financial reports?
- A. Motivation
 - B. Opportunity
 - C. Rationalization
16. A company is experiencing a period of strong financial performance. To increase the likelihood of exceeding analysts' earnings forecasts in the next reporting period, the company would *most likely* undertake accounting choices for the period under review that:
- A. inflate reported revenue.
 - B. delay expense recognition.
 - C. accelerate expense recognition.
17. Which of the following situations represents a motivation, rather than an opportunity, to issue low-quality financial reports?
- A. Poor internal controls
 - B. Search for a personal bonus
 - C. Inattentive board of directors
18. Which of the following situations will *most likely* motivate managers to inflate reported earnings?
- A. Possibility of bond covenant violation
 - B. Earnings that have exceeded analysts' forecasts
 - C. Earnings that have grown from the prior-year period
19. Which of the following *best* describes an opportunity for management to issue low-quality financial reports?
- A. Ineffective board of directors
 - B. Pressure to achieve some performance level
 - C. Corporate concerns about financing in the future
20. An audit opinion of a company's financial reports is *most likely* intended to:
- A. detect fraud.
 - B. reveal misstatements.
 - C. ensure that financial information is presented fairly.
21. If a company uses a non-GAAP financial measure in an SEC filing, then the company must:
- A. give more prominence to the non-GAAP measure if it is used in earnings releases.

- B. provide a reconciliation of the non-GAAP measure and equivalent GAAP measure.
 - C. exclude charges requiring cash settlement from any non-GAAP liquidity measures.
22. A company wishing to increase earnings in the reporting period may choose to:
- A. decrease the useful life of depreciable assets.
 - B. lower estimates of uncollectible accounts receivables.
 - C. classify a purchase as an expense rather than a capital expenditure.
23. Which technique *most likely* increases the cash flow provided by operations?
- A. Stretching the accounts payable credit period
 - B. Applying all non-cash discount amortization against interest capitalized
 - C. Shifting classification of interest paid from financing to operating cash flows
24. Bias in revenue recognition would *least likely* be suspected if:
- A. the firm engages in barter transactions.
 - B. reported revenue is higher than the previous quarter.
 - C. revenue is recognized before goods are shipped to customers.
25. Which of the following is an indication that a company may be recognizing revenue prematurely? Relative to its competitors, the company's:
- A. asset turnover is decreasing.
 - B. receivables turnover is increasing.
 - C. days sales outstanding is increasing.
26. Which of the following would *most likely* signal that a company may be using aggressive accrual accounting policies to shift current expenses to later periods? Over the last five-year period, the ratio of cash flow to net income has:
- A. increased each year.
 - B. decreased each year.
 - C. fluctuated from year to year.
27. An analyst reviewing a firm with a large reported restructuring charge to earnings should:
- A. view expenses reported in prior years as overstated.
 - B. disregard it because it is solely related to past events.
 - C. consider making pro forma adjustments to prior years' earnings.

SOLUTIONS

1. B is correct. Financial reporting quality pertains to the quality of information in financial reports. High-quality financial reporting provides decision-useful information, which is relevant and faithfully represents the economic reality of the company's activities. Earnings of high quality are sustainable and provide an adequate level of return. Highest-quality financial reports reflect both high financial reporting quality and high earnings quality.
2. C is correct. Financial reporting quality pertains to the quality of the information contained in financial reports. High-quality financial reports provide decision-useful information that faithfully represents the economic reality of the company. Low-quality financial reports impede assessment of earnings quality. Financial reporting quality is distinguishable from earnings quality, which pertains to the earnings and cash generated by the company's actual economic activities and the resulting financial condition. Low-quality earnings are not sustainable and decrease company value.
3. B is correct. Financial reporting quality pertains to the quality of the information contained in financial reports. If financial reporting quality is low, the information provided is of little use in assessing the company's performance. Financial reporting quality is distinguishable from earnings quality, which pertains to the earnings and cash generated by the company's actual economic activities and the resulting financial condition.
4. B is correct. Earnings quality pertains to the earnings and cash generated by the company's actual economic activities and the resulting financial condition. Low-quality earnings are likely not sustainable over time because the company does not expect to generate the same level of earnings in the future or because earnings will not generate sufficient return on investment to sustain the company. Earnings that are not sustainable decrease company value. Earnings quality is distinguishable from financial reporting quality, which pertains to the quality of the information contained in financial reports.
5. A is correct. Earnings that result from non-recurring activities are unsustainable. Unsustainable earnings are an example of lower-quality earnings. Recognizing earnings that result from non-recurring activities is neither a biased accounting choice nor indicative of lower quality financial reporting because it faithfully represents economic events.
6. B is correct. At the top of the quality spectrum of financial reports are reports that conform to GAAP, are decision useful, and have earnings that are sustainable and offer adequate returns. In other words, these reports have both high financial reporting quality and high earnings quality.
7. A is correct. Financial reports span a quality continuum from high to low based on decision-usefulness and earnings quality (see Exhibit 2). The lowest-quality reports portray fictitious events, which may misrepresent the company's performance or obscure fraudulent misappropriation of the company's assets.
8. B is correct. Deferring R&D investments into the next reporting period is an example of earnings management by taking a *real* action.
9. B is correct. High-quality financial reports offer useful information, meaning information that is relevant and faithfully represents actual performance. Although

low earnings quality may not be desirable, if the reported earnings are representative of actual performance, they are consistent with high-quality financial reporting. Highest-quality financial reports reflect both high financial reporting quality and high earnings quality.

10. B is correct. Aggressive accounting choices aim to enhance the company's reported performance by inflating the amount of revenues, earnings, or operating cash flow reported in the period. Consequently, the financial performance for that period would most likely exhibit an upward bias.
11. C is correct. Accounting choices are considered conservative if they decrease the company's reported performance and financial position in the current period under review. Conservative choices may increase the amount of debt reported on the balance sheet. They may decrease the revenues, earnings, or operating cash flow reported for the period and increase those amounts in later periods.
12. A is correct. Conservatism reduces the possibility of litigation and, by extension, litigation costs. Rarely, if ever, is a company sued because it understated good news or overstated bad news. Accounting conservatism is a type of bias in financial reporting that decreases a company's reported performance. Conservatism directly conflicts with the characteristic of neutrality.
13. B is correct. Managers often have incentives to meet or beat market expectations, particularly if management compensation is linked to increases in share prices or to reported earnings.
14. C is correct. Managers may be motivated to understate earnings in a period with unexpected strong performance by delaying revenue recognition or accelerating expense recognition to increase the probability of exceeding expectations in a subsequent period (referred to as "banking" some earnings for the next period.)
15. C is correct. Typically, conditions of opportunity, motivation, and rationalization exist when individuals issue low-quality financial reports. Rationalization occurs when an individual is concerned about a choice and needs to be able to justify it to herself or himself. If the manager is concerned about a choice in a financial report, the manager may ask for other opinions to convince herself or himself that it is okay.
16. C is correct. In a period of strong financial performance, managers may pursue accounting choices that increase the probability of exceeding earnings forecasts for the next period. By accelerating expense recognition or delaying revenue recognition, managers may reduce financial performance in the current period in order to inflate earnings in the next period and increase the likelihood of exceeding targets.
17. B is correct. Motivation can result from pressure to meet some criteria for personal reasons, such as a bonus, or corporate reasons, such as concern about future financing. Poor internal controls and an inattentive board of directors offer opportunities to issue low-quality financial reports.
18. A is correct. The possibility of bond covenant violations may motivate managers to inflate earnings in the reporting period. In so doing, the company may be able to avoid the consequences associated with violating bond covenants.
19. A is correct. Opportunities to issue low-quality financial reports include internal conditions, such as an ineffective board of directors, and external conditions, such as accounting standards that provide scope for divergent choices. Pressure

to achieve a certain level of performance and corporate concerns about future financing are examples of motivations to issue low-quality financial reports. Typically, three conditions exist when low-quality financial reports are issued: opportunity, motivation, and rationalization.

20. C is correct. An audit is intended to provide assurance that the company's financial reports are presented fairly, thus providing discipline regarding financial reporting quality. Regulatory agencies usually require that the financial statements of publicly traded companies be audited by an independent auditor to provide assurance that the financial statements conform to accounting standards. Privately held companies may also choose to obtain audit opinions either voluntarily or because an outside party requires it. An audit is not typically intended to detect fraud. An audit is based on sampling and it is possible that the sample might not reveal misstatements.
21. B is correct. If a company uses a non-GAAP financial measure in an SEC filing, it is required to provide the most directly comparable GAAP measure with equivalent prominence in the filing. In addition, the company is required to provide a reconciliation between the non-GAAP measure and the equivalent GAAP measure. Similarly, IFRS requires that any non-IFRS measures included in financial reports must be defined and their potential relevance explained. The non-IFRS measures must be reconciled with IFRS measures.
22. B is correct. If a company wants to increase reported earnings, the company's managers may reduce the allowance for uncollected accounts and the related expense reported for the period. Decreasing the useful life of depreciable assets would increase depreciation expense and decrease earnings in the reporting period. Classifying a purchase as an expense, rather than capital expenditure, would decrease earnings in the reporting period. The use of accrual accounting may result in estimates in financial reports, because all facts associated with events may not be known at the time of recognition. These estimates can be grounded in reality or managed by the company to present a desired financial picture.
23. A is correct. Managers can temporarily show a higher cash flow from operations by stretching the accounts payable credit period. In other words, the managers delay payments until the next accounting period. Applying all non-cash discount amortization against interest capitalized causes reported interest expenses and operating cash outflow to be higher, resulting in a lower cash flow provided by operations. Shifting the classification of interest paid from financing to operating cash flows lowers the cash flow provided by operations.
24. B is correct. Bias in revenue recognition can lead to manipulation of information presented in financial reports. Addressing the question as to whether revenue is higher or lower than the previous period is insufficient to determine if there is bias in revenue recognition. Additional analytical procedures must be performed to identify warning signals of accounting malfeasance. Barter transactions are difficult to value properly and may result in bias in revenue recognition. Policies that make it easier to prematurely recognize revenue, such as before goods are shipped to customers, may be a warning sign of accounting malfeasance.
25. C is correct. If a company's days sales outstanding (DSO) is increasing relative to competitors, this may be a signal that revenues are being recorded prematurely or are even fictitious. Numerous analytical procedures can be performed to provide evidence of manipulation of information in financial reporting. These warning signs are often linked to bias associated with revenue recognition and expense recognition policies.

26. B is correct. If the ratio of cash flow to net income for a company is consistently below 1 or has declined repeatedly over time, this may be a signal of manipulation of information in financial reports through aggressive accrual accounting policies. When net income is consistently higher than cash provided by operations, one possible explanation is that the company may be using aggressive accrual accounting policies to shift current expenses to later periods.
27. C is correct. To extrapolate historical earnings trends, an analyst should consider making pro forma analytical adjustments of prior years' earnings to reflect in those prior years a reasonable share of the current period's restructuring and impairment charges.

PRACTICE PROBLEMS

1. Comparison of a company's financial results to other peer companies for the same time period is called:
 - A. time-series analysis.
 - B. common-size analysis.
 - C. cross-sectional analysis.
2. An analyst observes a decrease in a company's inventory turnover. Which of the following would *most likely* explain this trend?
 - A. The company installed a new inventory management system, allowing more efficient inventory management.
 - B. Due to problems with obsolescent inventory last year, the company wrote off a large amount of its inventory at the beginning of the period.
 - C. The company installed a new inventory management system but experienced some operational difficulties resulting in duplicate orders being placed with suppliers.
3. Which of the following would *best* explain an increase in receivables turnover?
 - A. The company adopted new credit policies last year and began offering credit to customers with weak credit histories.
 - B. Due to problems with an error in its old credit scoring system, the company had accumulated a substantial amount of uncollectible accounts and wrote off a large amount of its receivables.
 - C. To match the terms offered by its closest competitor, the company adopted new payment terms now requiring net payment within 30 days rather than 15 days, which had been its previous requirement.
4. Brown Corporation had average days of sales outstanding of 19 days in the most recent fiscal year. Brown wants to improve its credit policies and collection practices and decrease its collection period in the next fiscal year to match the industry average of 15 days. Credit sales in the most recent fiscal year were \$300 million, and Brown expects credit sales to increase to \$390 million in the next fiscal year. To achieve Brown's goal of decreasing the collection period, the change in the average accounts receivable balance that must occur is *closest* to:
 - A. +USD0.41 million.
 - B. -USD0.41 million.
 - C. -USD1.22 million.

The following information relates to questions 5-5

An analyst is interested in assessing both the efficiency and liquidity of Spherion PLC. The analyst has collected the data in Exhibit 1 for Spherion:

Exhibit 1: Spherion Data

	FY3	FY2	FY1
Days of inventory on hand	32	34	40
Days sales outstanding	28	25	23
Number of days of payables	40	35	35

5. Based on the data in Exhibit 1, what is the analyst *least likely* to conclude?
 - A. Inventory management has contributed to improved liquidity.
 - B. Management of payables has contributed to improved liquidity.
 - C. Management of receivables has contributed to improved liquidity.
6. To assess a company's ability to fulfill its long-term obligations, an analyst would *most likely* examine:
 - A. activity ratios.
 - B. liquidity ratios.
 - C. solvency ratios.

The following information relates to questions 7-8

An analyst is evaluating the solvency and liquidity of Apex Manufacturing and has collected the data in Exhibit 1:

Exhibit 1: Solvency and Liquidity of Apex Manufacturing (euro millions)

	FY5	FY4	FY3
Total debt	2,000	1,900	1,750
Total equity	4,000	4,500	5,000

7. Which of the following would be the analyst's *most likely* conclusion?
 - A. The company is becoming less liquid, as evidenced by the increase in its debt-to-equity ratio from 0.35 to 0.50 from FY3 to FY5.

- B. The company is becoming increasingly more liquid, as evidenced by the increase in its debt-to-equity ratio from 0.35 to 0.50 from FY3 to FY5.
 - C. The company is becoming increasingly less solvent, as evidenced by the increase in its debt-to-equity ratio from 0.35 to 0.50 from FY3 to FY5.
8. What would be the *most* reasonable explanation of the financial data?
- A. The decline in the company's equity results from a decline in the market value of this company's common shares.
 - B. The EUR250 increase in the company's debt from FY3 to FY5 indicates that lenders are viewing the company as increasingly creditworthy.
 - C. The decline in the company's equity indicates that the company may be incurring losses, paying dividends greater than income, or repurchasing shares.
-
9. An analyst observes the data in Exhibit 1 for two companies:

Exhibit 1: Data Comparison (US dollars)

	Company A	Company B
Revenue	4,500	6,000
Net income	50	1,000
Current assets	40,000	60,000
Total assets	100,000	700,000
Current liabilities	10,000	50,000
Total debt	60,000	150,000
Shareholders' equity	30,000	500,000

Which of the following choices *best* describes reasonable conclusions that the analyst might make about the two companies' ability to pay their current and long-term obligations?

- A. Company A's current ratio of 4.0 indicates it is more liquid than Company B, whose current ratio is only 1.2, but Company B is more solvent, as indicated by its lower debt-to-equity ratio.
- B. Company A's current ratio of 0.25 indicates it is less liquid than Company B, whose current ratio is 0.83, and Company A is also less solvent, as indicated by a debt-to-equity ratio of 200 percent compared with Company B's debt-to-equity ratio of only 30 percent.
- C. Company A's current ratio of 4.0 indicates it is more liquid than Company B, whose current ratio is only 1.2, and Company A is also more solvent, as indicated by a debt-to-equity ratio of 200 percent compared with Company B's debt-to-equity ratio of only 30 percent.

The following information relates to questions 10-13

The following data appear in the five-year summary of a major international company. A business combination with another major manufacturer took place in FY13.

Exhibit 1: Five-Year Summary of a Major International Company

	FY10	FY11	FY12	FY13	FY14
Financial statements	GBP millions	GBP millions	GBP millions	GBP millions	GBP millions
Income statements					
Revenue	4,390	3,624	3,717	8,167	11,366
Profit before interest and taxation (EBIT)	844	700	704	933	1,579
Net interest expense	−80	−54	−98	−163	−188
Taxation	−186	−195	−208	−349	−579
Minorities	−94	−99	−105	−125	−167
Profit for the year	484	352	293	296	645
Balance sheets					
Fixed assets	3,510	3,667	4,758	10,431	11,483
Current asset investments, cash at bank and in hand	316	218	290	561	682
Other current assets	558	514	643	1,258	1,634
Total assets	4,384	4,399	5,691	12,250	13,799
Interest bearing debt (long term)	−602	−1,053	−1,535	−3,523	−3,707
Other creditors and provisions (current)	−1,223	−1,054	−1,102	−2,377	−3,108
Total liabilities	−1,825	−2,107	−2,637	−5,900	−6,815
Net assets	2,559	2,292	3,054	6,350	6,984
Shareholders' funds	2,161	2,006	2,309	5,572	6,165
Equity minority interests	398	286	745	778	819
Capital employed	2,559	2,292	3,054	6,350	6,984
Cash flow					

	FY10	FY11	FY12	FY13	FY14
Working capital movements	-53	5	71	85	107
Net cash inflow from operating activities	864	859	975	1,568	2,292

10. The company's total assets at year-end FY9 were GBP3,500 million. Which of the following choices *best* describes reasonable conclusions an analyst might make about the company's efficiency?
- A. Comparing FY14 with FY10, the company's efficiency deteriorated, as indicated by its current ratio.
 - B. Comparing FY14 with FY10, the company's efficiency deteriorated due to asset growth faster than turnover revenue growth.
 - C. Comparing FY14 with FY10, the company's efficiency improved, as indicated by a total asset turnover ratio of 0.86 compared with 0.64.
11. Which of the following choices *best* describes reasonable conclusions an analyst might make about the company's solvency?
- A. Comparing FY14 with FY10, the company's solvency improved, as indicated by the growth in its profits to GBP 645 million.
 - B. Comparing FY14 with FY10, the company's solvency deteriorated, as indicated by a decrease in interest coverage from 10.6 to 8.4.
 - C. Comparing FY14 with FY10, the company's solvency improved, as indicated by an increase in its debt-to-assets ratio from 0.14 to 0.27.
12. Which of the following choices *best* describes reasonable conclusions an analyst might make about the company's liquidity?
- A. Comparing FY14 with FY10, the company's liquidity improved, as indicated by an increase in its current ratio from 0.71 to 0.75.
 - B. Comparing FY14 with FY10, the company's liquidity deteriorated, as indicated by a decrease in interest coverage from 10.6 to 8.4.
 - C. Comparing FY14 with FY10, the company's liquidity improved, as indicated by an increase in its debt-to-assets ratio from 0.14 to 0.27.
13. Which of the following choices *best* describes reasonable conclusions an analyst might make about the company's profitability?
- A. Comparing FY14 with FY10, the company's profitability improved, as indicated by an increase in its debt-to-assets ratio from 0.14 to 0.27.
 - B. Comparing FY14 with FY10, the company's profitability improved, as indicated by the growth in its shareholders' equity to GBP6,165 million.
 - C. Comparing FY14 with FY10, the company's profitability deteriorated, as indicated by a decrease in its net profit margin from 11.0 percent to 5.7 percent.

14. An analyst compiles the data in Exhibit 1 for a company:

Exhibit 1: Net Profit Margin			
	FY13	FY14	FY15
ROE	19.8%	20.0%	22.0%
Return on total assets	8.1%	8.0%	7.9%
Total asset turnover	2.0	2.0	2.1

Based only on the information above, the *most* appropriate conclusion is that, over the period FY13 to FY15, the company's:

- A. net profit margin and financial leverage have decreased.
- B. net profit margin and financial leverage have increased.
- C. net profit margin has decreased but its financial leverage has increased.

15. A decomposition of ROE for Integra SA is as follows:

Exhibit 1: Integra SA ROE		
	FY12	FY11
ROE	18.90%	18.90%
Tax burden	0.70	0.75
Interest burden	0.90	0.90
EBIT margin	10.00%	10.00%
Asset turnover	1.50	1.40
Leverage	2.00	2.00

Which of the following choices *best* describes reasonable conclusions an analyst might make based on this ROE decomposition?

- A. Profitability and the liquidity position both improved in FY12.
- B. The higher average tax rate in FY12 offset the improvement in profitability, leaving ROE unchanged.
- C. The higher average tax rate in FY12 offset the improvement in efficiency, leaving ROE unchanged.

16. A decomposition of ROE for Company A and Company B is as follows:

Exhibit 1: ROE for Company A and Company B

	Company A		Company B	
	FY15	FY14	FY15	FY14
ROE	26.46%	18.90%	26.33%	18.90%
Tax burden	0.7	0.75	0.75	0.75
Interest burden	0.9	0.9	0.9	0.9
EBIT margin	7.00%	10.00%	13.00%	10.00%
Asset turnover	1.5	1.4	1.5	1.4
Leverage	4	2	2	2

An analyst is *most likely* to conclude that:

- A. Company A's ROE is higher than Company B's in FY15, and one explanation consistent with the data is that Company A may have purchased new, more efficient equipment.
- B. the difference between the two companies' ROE in FY15 is very small and Company A's ROE remains similar to Company B's ROE mainly due to Company A increasing its financial leverage.
- C. Company A's ROE is higher than Company B's in FY15, and one explanation consistent with the data is that Company A has made a strategic shift to a product mix with higher profit margins.

17. When developing forecasts, analysts should *most likely*:

- A. develop possibilities relying exclusively on the results of financial analysis.
- B. aim to develop extremely precise forecasts using the results of financial analysis.
- C. use the results of financial analysis, analysis of other information, and judgment.

SOLUTIONS

1. C is correct. Cross-sectional analysis involves the comparison of companies with each other for the same time period. Time-series or trend analysis is the comparison of financial data across different time periods. Common-size analysis involves expressing financial data in relation to a single financial statement item, or base.
2. C is correct. The company's problems with its inventory management system causing duplicate orders would likely result in a higher amount of inventory and, therefore, would result in a decrease in inventory turnover. A more efficient inventory management system and a write-off of inventory at the beginning of the period would both likely decrease the average inventory for the period (the denominator of the inventory turnover ratio), thus increasing the ratio rather than decreasing it.
3. B is correct. A write-off of receivables would decrease the average amount of accounts receivable (the denominator of the receivables turnover ratio), thus increasing this ratio. Customers with weaker credit are more likely to make payments more slowly or to pose collection difficulties, which would likely increase the average amount of accounts receivable and thus decrease receivables turnover. Longer payment terms would likely increase the average amount of accounts receivable and thus decrease receivables turnover.
4. A is correct. The average accounts receivable balances (actual and desired) must be calculated to determine the desired change. The average accounts receivable balance can be calculated as an average day's credit sales times the DSO. For the most recent fiscal year, the average accounts receivable balance is USD15.62 million $[= (\text{USD}300,000,000/365) \times 19]$. The desired average accounts receivable balance for the next fiscal year is USD16.03 million $[= (\text{USD}390,000,000/365) \times 15]$. This is an increase of USD0.41 million $[= 16.03 \text{ million} - 15.62 \text{ million}]$. An alternative approach is to calculate the turnover and divide sales by turnover to determine the average accounts receivable balance. Turnover equals 365 divided by DSO. Turnover is 19.21 $[= 365/19]$ for the most recent fiscal year and is targeted to be 24.33 $[= 365/15]$ for the next fiscal year. The average accounts receivable balances are USD15.62 million $[= \text{USD}300,000,000/19.21]$, and USD16.03 million $[= \text{USD}390,000,000/24.33]$. The change is an increase in receivables of USD0.41 million.
5. C is correct. The analyst is *unlikely* to reach the conclusion given in Statement C because days of sales outstanding increased from 23 days in FY1 to 25 days in FY2 to 28 days in FY3, indicating that the time required to collect receivables has increased over the period. This is a negative factor for Spherion's liquidity. By contrast, days of inventory on hand dropped over the period FY1 to FY3, a positive for liquidity. The company's increase in days payable, from 35 days to 40 days, shortened its cash conversion cycle, thus also contributing to improved liquidity.
6. C is correct. Solvency ratios are used to evaluate the ability of a company to meet its long-term obligations. An analyst is more likely to use activity ratios to evaluate how efficiently a company uses its assets. An analyst is more likely to use liquidity ratios to evaluate the ability of a company to meet its short-term obligations.
7. C is correct. The company is becoming increasingly less solvent, as evidenced by its debt-to-equity ratio increasing from 0.35 to 0.50 from FY3 to FY5. The

amount of a company's debt and equity do not provide direct information about the company's liquidity position.

Debt to equity:

$$\text{FY5: } 2,000/4,000 = 0.5000$$

$$\text{FY4: } 1,900/4,500 = 0.4222$$

$$\text{FY3: } 1,750/5,000 = 0.3500$$

8. C is correct. The decline in the company's equity indicates that the company may be incurring losses, paying dividends greater than income, or repurchasing shares. Recall that Beginning equity – Shares repurchased + Comprehensive income – Dividends = Ending equity. The book value of a company's equity is not affected by changes in the market value of its common stock. An increased amount of lending does not necessarily indicate that lenders view a company as increasingly creditworthy. Creditworthiness is not evaluated based on how much a company has increased its debt but rather on its willingness and ability to pay its obligations. (Its financial strength is indicated by its solvency, liquidity, profitability, efficiency, and other aspects of credit analysis.)
9. A is correct. Company A's current ratio of 4.0 (= USD40,000/USD10,000) indicates it is more liquid than Company B, whose current ratio is only 1.2 (=USD60,000/USD50,000). Company B is more solvent, as indicated by its lower debt-to-equity ratio of 30 percent (= USD150,000/USD500,000) compared with Company A's debt-to-equity ratio of 200 percent (= USD60,000/USD30,000).
10. B is correct. The company's efficiency deteriorated, as indicated by the decline in its total asset turnover ratio from 1.11 {= 4,390/[(4,384 + 3,500)/2]} for FY10 to 0.87 {= 11,366/[(12,250 + 13,799)/2]} for FY14. The decline in the total asset turnover ratio resulted from an increase in average total assets from GBP3,942 [= (4,384 + 3,500)/2] for FY10 to GBP13,024.5 for FY14, an increase of 230 percent, compared with an increase in revenue from GBP4,390 in FY10 to GBP11,366 in FY14, an increase of only 159 percent. The current ratio is not an indicator of efficiency.
11. B is correct. Comparing FY14 with FY10, the company's solvency deteriorated, as indicated by a decrease in interest coverage from 10.6 (= 844/80) in FY10 to 8.4 (= 1,579/188) in FY14. The debt-to-asset ratio increased from 0.14 (= 602/4,384) in FY10 to 0.27 (= 3,707/13,799) in FY14. This is also indicative of deteriorating solvency. In isolation, the amount of profits does not provide enough information to assess solvency.
12. A is correct. Comparing FY14 with FY10, the company's liquidity improved, as indicated by an increase in its current ratio from 0.71 [= (316 + 558)/1,223] in FY10 to 0.75 [= (682 + 1,634)/3,108] in FY14. Note, however, comparing only the cash ratio shows a decline in liquidity from 0.26 (= 316/1,223) in FY10 to 0.22 (= 682/3,108) in FY14. Debt-to-assets ratio and interest coverage are measures of solvency not liquidity.
13. C is correct. Comparing FY14 with FY10, the company's profitability deteriorated, as indicated by a decrease in its net profit margin from 11.0 percent (= 484/4,390) to 5.7 percent (= 645/11,366). Debt-to-assets ratio is a measure of solvency not an indicator of profitability. Growth in shareholders' equity, in isolation, does not provide enough information to assess profitability.
14. C is correct. The company's net profit margin has decreased and its financial

leverage has increased. $ROA = \text{Net profit margin} \times \text{Total asset turnover}$. ROA decreased over the period despite the increase in total asset turnover; therefore, the net profit margin must have decreased.

$ROE = \text{Return on assets} \times \text{Financial leverage}$. ROE increased over the period despite the drop in ROA; therefore, financial leverage must have increased.

15. C is correct. The increase in the average tax rate in FY12, as indicated by the decrease in the value of the tax burden (the tax burden equals one minus the average tax rate), offset the improvement in efficiency indicated by higher asset turnover) leaving ROE unchanged. The EBIT margin, measuring profitability, was unchanged in FY12 and no information is given on liquidity.
16. B is correct. The difference between the two companies' ROE in FY15 is very small and is mainly the result of Company A's increase in its financial leverage, indicated by the increase in its Assets/Equity ratio from 2 to 4. The impact of efficiency on ROE is identical for the two companies, as indicated by both companies' asset turnover ratios of 1.5. Furthermore, if Company A had purchased newer equipment to replace older, depreciated equipment, then the company's asset turnover ratio (computed as sales/assets) would have declined, assuming constant sales. Company A has experienced a significant decline in its operating margin, from 10 percent to 7 percent which, all else equal, would not suggest that it is selling more products with higher profit margins.
17. C is correct. The results of an analyst's financial analysis are integral to the process of developing forecasts, along with the analysis of other information and judgment of the analysts. Forecasts are not limited to a single point estimate but should involve a range of possibilities.

PRACTICE PROBLEMS

The following information relates to questions 1-7

Nigel French, an analyst at Taurus Investment Management, is analyzing Archway Technologies, a manufacturer of luxury electronic auto equipment, at the request of his supervisor, Lukas Wright. French is asked to evaluate Archway's profitability over the past five years relative to its two main competitors, which are located in different countries with significantly different tax structures.

French begins by assessing Archway's competitive position within the luxury electronic auto equipment industry using Porter's five forces framework. A summary of French's industry analysis is presented in Exhibit 1.

Exhibit 1: Analysis of Luxury Electronic Auto Equipment Industry Using Porter's Five Forces Framework

Force	Factors to Consider
Threat of substitutes	Customer switching costs are high
Rivalry	Archway holds 60 percent of world market share; each of its two main competitors holds 15 percent
Bargaining power of suppliers	Primary inputs are considered basic commodities, and there are a large number of suppliers
Bargaining power of buyers	Luxury electronic auto equipment is very specialized (non-standardized)
Threat of new entrants	High fixed costs to enter industry

French notes that for the year just ended (2019), Archway's COGS was 30 percent of sales. To forecast Archway's income statement for 2020, French assumes that all companies in the industry will experience an inflation rate of 8 percent on the COGS. Exhibit 2 shows French's forecasts relating to Archway's price and volume changes.

Exhibit 2: Archway's 2020 Forecasted Price and Volume Changes

Average price increase per unit	5.00%
Volume growth	-3.00%

After putting together income statement projections for Archway, French forecasts Archway's balance sheet items. He uses Archway's historical efficiency ratios to forecast the company's working capital accounts.

Based on his financial forecast for Archway, French estimates a terminal value using a valuation multiple based on the company's average price-to-earnings multiple (P/E) over the past five years. Wright discusses with French how the terminal value estimate is sensitive to key assumptions about the company's future

prospects. Wright asks French:

“What change in the calculation of the terminal value would you make if a technological development that would adversely affect Archway was forecast to occur sometime beyond your financial forecast horizon?”

1. Which profitability metric should French use to assess Archway’s five-year historic performance relative to its competitors?
 - A. Current ratio
 - B. Operating margin
 - C. Return on invested capital
2. Based on the current competitive landscape presented in Exhibit 1, French should conclude that Archway’s ability to:
 - A. pass along price increases is high.
 - B. demand lower input prices from suppliers is low.
 - C. generate above-average returns on invested capital is low.
3. Based on the current competitive landscape presented in Exhibit 1, Archway’s operating profit margins over the forecast horizon are *least likely* to:
 - A. decrease.
 - B. remain constant.
 - C. increase.
4. Based on Exhibit 2, Archway’s forecasted gross profit margin for 2020 is *closest* to:
 - A. 62.7 percent.
 - B. 67.0 percent.
 - C. 69.1 percent.
5. French’s approach to forecasting Archway’s working capital accounts would be *most likely* classified as a:
 - A. hybrid approach.
 - B. top-down approach.
 - C. bottom-up approach.
6. The *most appropriate* response to Wright’s question about the technological development is to:
 - A. increase the required return.
 - B. decrease the perpetual growth rate.
 - C. decrease the price-to-earnings multiple.
7. If the luxury electronic auto equipment industry is subject to rapid technological changes and market share shifts, how should French *best* adapt his approach to

modeling?

- A. Examine base rates
 - B. Forecast multiple scenarios
 - C. Speak to analysts who hold diverse opinions on the stock
-

The following information relates to questions 8-14

Gertrude Fromm is a transportation sector analyst at Tucana Investments. She is conducting an analysis of Omikroon, N.V., a hypothetical European engineering company that manufactures and sells scooters and commercial trucks.

Omikroon's petrol scooter division is the market leader in its sector and has two competitors. Omikroon's petrol scooters have a strong brand name and a well-established distribution network. Given the strong branding established by the market leaders, the cost of entering the industry is high. But Fromm anticipates that small, inexpensive, imported petrol-fueled motorcycles could become substitutes for Omikroon's petrol scooters.

Fromm uses ROIC as the metric to assess Omikroon's performance.

Omikroon has just introduced the first electric scooter to the market at year-end 2019. The company's expectations are as follows:

- Competing electric scooters will reach the market in 2021.
- Electric scooters will not be a substitute for petrol scooters.
- The important research costs in 2020 and 2021 will lead to more efficient electric scooters.

Fromm decides to use a five-year forecast horizon for Omikroon after considering the following three factors:

- Factor 1 The annual portfolio turnover at Tucana Investments is 30 percent.
- Factor 2 The electronic scooter industry is expected to grow rapidly over the next 10 years.
- Factor 3 Omikroon has announced it would acquire a light truck manufacturer that will be fully integrated into its truck division by 2021 and will add 2 percent to the company's total revenues.

Fromm uses the base case forecast for 2020 shown in Exhibit 1 to perform the following sensitivity analysis:

- The price of an imported specialty metal used for engine parts increases by 20 percent.
- This metal constitutes 4 percent of Omikroon's cost of sales.
- Omikroon will not be able to pass on the higher metal expense to its customers.

Exhibit 1: Omikroon's Selected Financial Forecasts for 2020 Base Case (euro millions)

	Petrol Scooter Division	Commercial Truck Division	Electric Scooter Division	Total
Sales	99.05	45.71	7.62	152.38
Cost of sales				105.38
Gross profit				47.00
Operating profit				9.20

Omikroon will initially outsource its electric scooter parts. But manufacturing these parts in-house beginning in 2021 will imply changes to an existing factory. This factory cost EUR7 million three years ago and had an estimated useful life of 10 years. Fromm is evaluating two scenarios:

Scenario 1 Refit the existing factory for EUR27 million.

Scenario 2 Sell the existing factory for EUR5 million. Build a new factory costing EUR30 million with a useful life of 10 years.

8. Using Porter's five forces analysis, which of the following competitive factors is *most likely* to have the greatest impact on Omikroon's petrol scooter pricing power?
 - A. Rivalry
 - B. Threat of substitutes
 - C. Threat of new entrants
9. The metric used by Fromm to assess Omikroon's performance incorporates:
 - A. the degree of financial leverage.
 - B. operating liabilities relative to operating assets.
 - C. the firm's competitiveness relative to companies in other tax regimes.
10. Based on Omikroon's expectations, the gross profit margin of Omikroon's electric scooter division in 2021 is *most likely* to be affected by:
 - A. competition.
 - B. research costs.
 - C. cannibalization by petrol scooters.
11. Which factor *best* justifies the five-year forecast horizon for Omikroon selected by Fromm?
 - A. Factor 1
 - B. Factor 2
 - C. Factor 3
12. Fromm's sensitivity analysis will result in a decrease in the 2020 base case gross

profit margin *closest to*:

- A. 0.55 percent.
 - B. 0.80 percent.
 - C. 3.32 percent.
13. Fromm's estimate of growth capital expenditures included in Omikroon's PP&E under Scenario 1 should be:
- A. lower than under Scenario 2.
 - B. the same as under Scenario 2.
 - C. higher than under Scenario 2.
14. To validate the forecast for rapid growth in the electronic scooter market over the next 10 years, Fromm speaks to the management of Omikroon and investor relations of ZeroWheel, a competitor. Which behavioral bias is Fromm *most likely* subject to?
- A. Confirmation
 - B. Conservatism
 - C. Overconfidence

The following information relates to questions 15-21

Angela Green, an investment manager at Horizon Investments, intends to hire a new investment analyst. After conducting initial interviews, Green has narrowed the pool to three candidates. She plans to conduct second interviews to further assess the candidates' knowledge of industry and company analysis.

Prior to the second interviews, Green asks the candidates to analyze Chrome Network Systems, a company that manufactures internet networking products. Each candidate is provided Chrome's financial information presented in Exhibit 1.

Exhibit 1: Chrome Network Systems Selected Financial Information (US dollar millions)

	Year-End		
	2017	2018	2019
Net sales	46.8	50.5	53.9
Cost of sales	18.2	18.4	18.8
Gross profit	28.6	32.1	35.1
SG&A expenses	19.3	22.5	25.1
Operating income	9.3	9.6	10.0
Interest expense	0.5	0.7	0.6

	Year-End		
	2017	2018	2019
Income before provision for income tax	8.8	8.9	9.4
Provision for income taxes	2.8	2.8	3.1
Net income	6.0	6.1	6.3

Green asks each candidate to forecast the 2020 income statement for Chrome and to outline the key assumptions used in their analysis. The job candidates are told to include Horizon's economic outlook for 2020 in their analysis, which assumes nominal GDP growth of 3.6 percent, based on expectations of real GDP growth of 1.6 percent and inflation of 2.0 percent.

Green receives the models from each of the candidates and schedules second interviews. To prepare for the interviews, Green compiles a summary of the candidates' key assumptions in Exhibit 2.

Exhibit 2: Summary of Key Assumptions Used in Candidates' Models

Metric	Candidate A	Candidate B	Candidate C
Net sales	Net sales will grow at the average annual growth rate in net sales over the 2017–19 time period.	Industry sales will grow at the same rate as nominal GDP, but Chrome will have a two-percentage-point decline in market share.	Net sales will grow 50 bps slower than nominal GDP.
Cost of sales	The 2020 gross margin will be the same as the average annual gross margin over the 2017–19 time period.	The 2020 gross margin will decline as costs increase by expected inflation.	The 2020 gross margin will increase by 20 bps from 2019.
SG&A expenses	The 2020 SG&A/net sales ratio will be the same as the average ratio over the 2017–19 time period.	The 2020 SG&A will grow at the rate of inflation.	The 2020 SG&A/net sales ratio will be the same as the 2019 ratio.
Interest expense	The 2020 interest expense assumes the effective interest rate will be the same as the 2019 rate.	The 2020 interest expense will be the same as the 2019 interest expense.	The 2020 interest expense will be the same as the average expense over the 2017–19 time period.
Income taxes	The 2020 effective tax rate will be the same as the 2019 rate.	The 2020 effective tax rate will equal the blended statutory rate of 30%.	The 2020 effective tax rate will be the same as the average effective tax rate over the 2017–19 time period.

15. Based on Exhibit 1, which of the following provides the strongest evidence that Chrome displays economies of scale?

- A. Increasing net sales
- B. Profit margins that are increasing with net sales

- C. Gross profit margins that are increasing with net sales
16. Based on Exhibit 2, the job candidate *most likely* using a bottom-up approach to model net sales is:
- A. Candidate A.
 - B. Candidate B.
 - C. Candidate C.
17. Based on Exhibit 2, the modeling approach used by Candidate B to project future net sales is *most accurately* classified as a:
- A. hybrid approach.
 - B. top-down approach.
 - C. bottom-up approach.
18. Based on Exhibits 1 and 2, Candidate C's forecast for cost of sales in 2020 is *closest* to:
- A. USD18.3 million.
 - B. USD18.9 million.
 - C. USD19.3 million.
19. Based on Exhibits 1 and 2, Candidate A's forecast for SG&A expenses in 2020 is *closest* to:
- A. USD23.8 million.
 - B. USD25.5 million.
 - C. USD27.4 million.
20. Based on Exhibit 2, forecasted interest expense will reflect changes in Chrome's debt level under the forecast assumptions used by:
- A. Candidate A.
 - B. Candidate B.
 - C. Candidate C.
21. Candidate B asks Green if she had additional information on Horizon's industry peers and competitors, to put the profitability estimates in a richer context. By asking for this additional information for their analysis, Candidate B is *most likely* seeking to mitigate which behavioral bias?
- A. Conservatism
 - B. Base rate neglect
 - C. illusion of control
-

SOLUTIONS

1. B is correct. Operating (EBIT) margin is a pre-tax profitability measure that can be useful in the peer comparison of companies in countries with different tax structures. Archway's two main competitors are located in different countries with significantly different tax structures; therefore, a pre-tax measure is better than an after-tax measure, such as ROIC. The current ratio is a liquidity measure, not a profitability measure.
2. A is correct. Porter's five forces framework in Exhibit 1 describes an industry with high barriers to entry, high customer switching costs (suggesting a low threat of substitutes), and a specialized product (suggesting low bargaining power of buyers). Furthermore, the primary production inputs from the large group of suppliers are considered basic commodities (suggesting low bargaining power of suppliers). These favorable industry characteristics will likely enable Archway to pass along price increases and generate above-average returns on invested capital.
3. A is correct. The current favorable characteristics of the industry (high barriers to entry, low bargaining power of suppliers and buyers, low threat of substitutes), coupled with Archway's dominant market share position, will likely lead to Archway's profit margins being at least equal to or greater than current levels over the forecast horizon.
4. C is correct. The calculation of Archway's gross profit margin for 2020, which reflects the industry-wide 8% inflation on COGS, is calculated as follows:

Revenue growth	1.85%
COGS increase	4.76%
Forecasted revenue (Base revenue = 100)	101.85
Forecasted COGS (Base COGS = 30)	31.43
Forecasted gross profit	70.42
Forecasted gross profit margin	69.14%

$$\begin{aligned}
 \text{Revenue growth} &= (1 + \text{Price increase for revenue}) \times (1 + \text{Volume growth}) - 1 \\
 &= (1.05) \times (0.97) - 1 \\
 &= 1.85\%.
 \end{aligned}$$

$$\begin{aligned}
 \text{COGS increase} &= (1 + \text{Price increase for COGS}) \times (1 + \text{Volume growth}) - 1 \\
 &= (1.08) \times (0.97) - 1 \\
 &= 4.76\%.
 \end{aligned}$$

$$\begin{aligned}
 \text{Forecasted revenue} &= \text{Base revenue} \times \text{Revenue growth increase} \\
 &= 100 \times 1.0185 \\
 &= 101.85.
 \end{aligned}$$

$$\text{Forecasted COGS} = \text{Base COGS} \times \text{COGS increase}$$

$$= 30 \times 1.0476$$

$$= 31.43.$$

$$\text{Forecasted gross profit} = \text{Forecasted revenue} - \text{Forecasted COGS}$$

$$= 101.85 - 31.43$$

$$= 70.42.$$

$$\text{Forecasted gross profit margin} = \text{Forecasted gross profit} / \text{Forecasted revenue}$$

$$= 70.42 / 101.85$$

$$= 69.14\%.$$

5. C is correct. French is using a bottom-up approach to forecast Archway's working capital accounts by using the company's historical efficiency ratios to project future performance.
6. C is correct. If the future growth or profitability of a company is likely to be lower than the historical average (in this case, because of a potential technological development), then the target multiple should reflect a discount to the historical multiple to reflect this difference in growth and/or profitability. If a multiple is used to derive the terminal value of a company, the choice of the multiple should be consistent with the long-run expectations for growth and required return. French tells Wright he believes that such a technological development could have an adverse impact on Archway beyond the forecast horizon.
7. B is correct. Forecasting a single scenario would not be appropriate given the high degree of uncertainty and range of potential outcomes for companies in this industry.
8. B is correct. Small, inexpensive, imported petrol-fueled motorcycles are substitutes for petrol scooters and could increasingly have an impact on Omikroon's petrol scooter pricing power.
9. B is correct. Return on invested capital is net operating profit minus adjusted taxes divided by invested capital, where invested capital is defined as operating assets minus operating liabilities.
10. A is correct. Competition from other electric scooter manufacturers is expected to begin in one year. After this time, competing electric scooters could lead to lower demand for Omikroon's electric scooters and affect Omikroon's gross profit margin.
11. B is correct. The electric scooter market is expected to grow rapidly, so the contribution of Omikroon's new electric scooter division is forecast to expand significantly over the next 10 years. A is incorrect because the investment company's portfolio turnover is not relevant for forecasting Omikroon's future results. C is incorrect because the light truck division is expected to add only 2% to total revenues in the future.
12. A is correct. The sensitivity analysis consists of an increase of 20 percent in the price of an input that constitutes 4 percent of cost of sales. Change in gross profit margin because of that increase is calculated as the change in cost of sales because of price increase divided by sales:

$$= (\text{Cost of sales} \times 0.04 \times 0.2) / \text{Sales}$$

$$= (105.38 \times 0.04 \times 0.2) / 152.38$$

$$= 0.0055 \text{ or } 0.55\%$$

13. C is correct. In Scenario 1, growth capital expenditures of EUR27 million for the refit of the existing idle factory is higher than the growth capital expenditures in Scenario 2 of EUR25 million. The EUR25 million is the cost of building a new factory for EUR30 million less the proceeds from the sale of the existing idle factory of EUR5 million.
14. A is correct. The management of Omikroon and investor relations of ZeroWheel are almost certainly biased in favor of expecting strong growth for the markets they participate in. To evaluate the forecast, Fromm should seek more independent sources and balance the biased sources with sources biased in the opposite direction or an analyst who is more skeptical.
15. C is correct. Economies of scale are a situation in which average costs decrease with increasing sales volume. Chrome's gross margins have been increasing with net sales. Gross margins that increase with sales levels provide evidence of economies of scale, assuming that higher levels of sales reflect increased unit sales. Gross margin more directly reflects the cost of sales than does profit margin.

Metric	2017	2018	2019
Net sales	\$46.8	\$50.5	\$53.9
Gross profit	28.6	32.1	35.1
Gross margin (gross profit/ net sales)	61.11%	63.56%	65.12%

16. A is correct. A bottom-up approach for developing inputs to equity valuation models begins at the level of the individual company or a unit within the company. By modeling net sales using the average annual growth rate, Candidate A is using a bottom-up approach. B and C are incorrect because both Candidate B and Candidate C are using a top-down approach, which begins at the level of the overall economy.
17. B is correct. A top-down approach usually begins at the level of the overall economy. Candidate B assumes industry sales will grow at the same rate as nominal GDP but that Chrome will have a 2-percentage-point decline in market share. A and C are incorrect because Candidate B is not using any elements of a bottom-up approach; therefore, a hybrid approach is not being employed.
18. C is correct. Candidate C assumes that the 2020 gross margin will increase by 20 bps from 2019 and that net sales will grow at 50 bps slower than nominal GDP (nominal GDP = Real GDP + Inflation = 1.6% + 2.0% = 3.6%). Accordingly, the 2020 forecasted cost of sales is USD19.27 million, rounded to USD19.3 million.

Metric	Calculation	Result
2020 gross margin = 2019 gm + 20 bps	USD35.1/USD53.9 = 65.12% + 0.20% =	65.32%
2020 CoS/net sales = 100% – gross margin	100% – 65.32% =	34.68%

Metric	Calculation	Result
2020 net sales = 2019 net sales × (1 + Nominal GDP – 0.50%)	USD53.9 million × (1 + 0.036 – 0.005) = USD53.9 million × 1.031 =	USD55.57 million
2020 cost of sales = 2020 net sales × CoS/net sales	USD55.57 × 34.68% =	USD19.27 million

19. B is correct. Candidate A assumes that the 2020 SG&A/net sales will be the same as the average SG&A/net sales over the 2017–19 time period and that net sales will grow at the annual average growth rate in net sales over the 2017–19 time period. Accordingly, the 2020 forecasted SG&A expenses are USD25.5 million.

Metric	Calculation	Result
Average SG&A/net sales, 2017–2019*	(41.24% + 44.55% + 46.57%)/3 =	44.12%
Average annual growth sales in net sales, 2017–2019**	(7.91% + 6.73%)/2 =	7.32%
2020 net sales = 2019 net sales × (1 + Average annual growth rate in net sales)	USD53.9 million × 1.0732 =	\$57.85 million
2020 SG&A = 2020 net sales × Average SG&A/net sales	USD57.85 million × 44.12% =	\$25.52 million

	2017	2018	2019
Net Sales	USD46.8	USD50.5	USD53.9
SG&A expenses	USD19.3	USD22.5	USD25.1
SG&A-to-sales ratio	41.24%	44.55%	46.57%

Year	Calculation
2018	(USD50.5/USD46.8) – 1 = 7.91%
2019	(USD53.9/USD50.5) – 1 = 6.73%

20. A is correct. In forecasting financing costs, such as interest expense, the debt/equity structure of a company is a key determinant. Accordingly, a method that recognizes the relationship between the income statement account (interest expense) and the balance sheet account (debt) would be a preferable method for forecasting interest expense when compared with methods that forecast based solely on the income statement account. By using the effective interest rate (interest expense divided by average gross debt), Candidate A is taking the debt/equity structure into account. B and C are incorrect because Candidate B (who forecasts 2020 interest expense to be the same as 2019 interest expense) and Candidate C (who forecasts 2020 interest expense to be the same as the 2017–19 average interest expense) are not taking the balance sheet into consideration.
21. B is correct. Base rates refer to attributes of a reference class and base rate neglect is ignoring such class information in favor of specific information. By incorporating industry data, Candidate B is seeking to mitigate this behavioral bias.